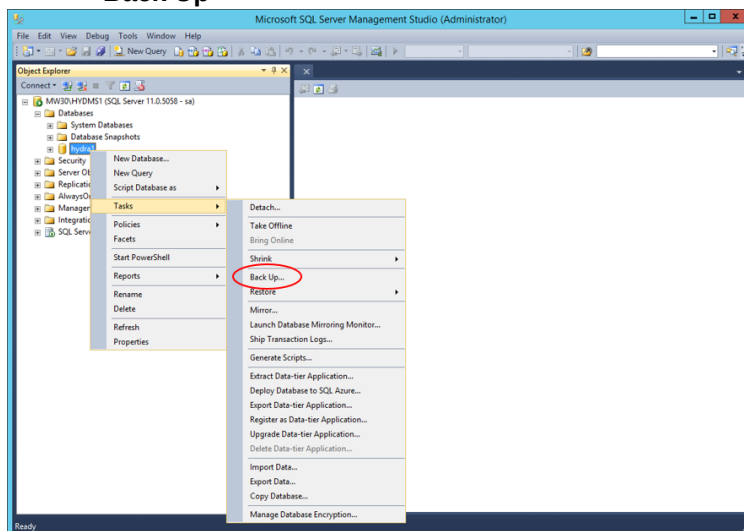


1 Database Backup (MS-SQL Server 2012)

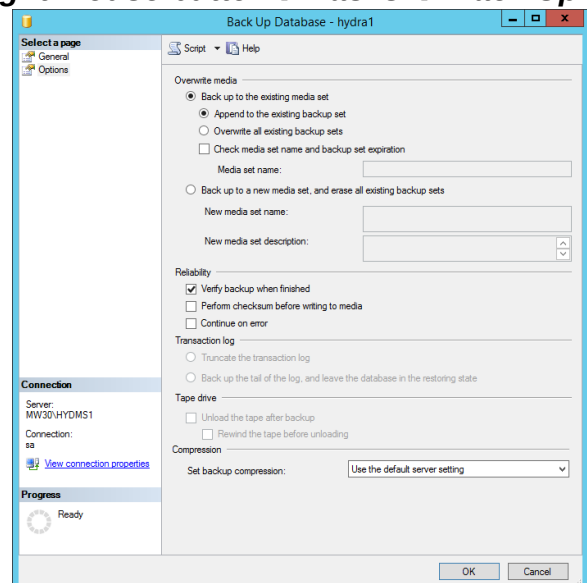
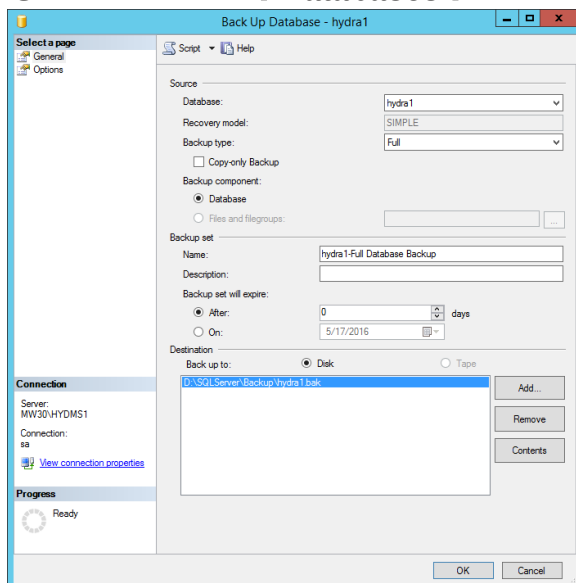
Manual database backup using SQL Server Management Studio

A backup of the database content is achieved by using the Microsoft SQL Server Management Studio. Procedure:

- Extend the SQL server instance (server name\HYDMS1) needing a backup in the Object Explorer, and then extend **Databases**.
- Then make a right mouse button click on the **HYDRA database**, click on **Tasks** and then on **Back Up**

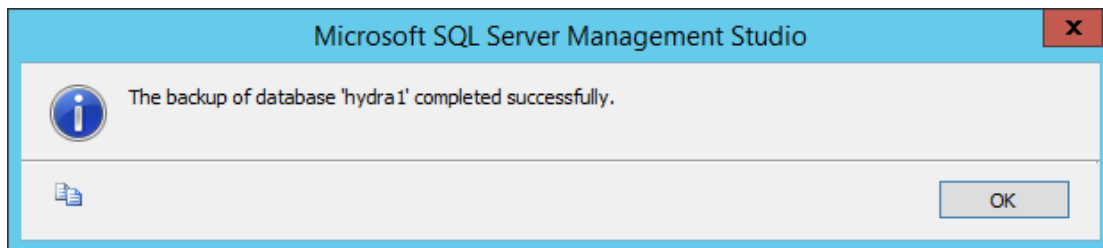


SERVER NAME → Databases → HYDRA1 → right mouse button → Tasks → Back Up



The backup type may be **complete** or **simple** according to the setting of the database.

Now you only have to specify the target, whether the database backup is to be made on the hard drive or on a tape drive. If no target has been defined yet, a backup target can be defined by adding it. The Options tab is used to specify how the backup is to be made. Upon confirming by pressing the OK button, the following message is displayed upon completion of the backup:



A data backup under SQL Server 2012 does not necessarily have to be made on a tape drive. If the database is configured appropriately, a backup in a file on the hard drive is also possible. Data backup on a local drive has advantages mainly in terms of performance, but it is also helpful as regards the recovery of a database. If possible, the backup should be implemented on a local hard drive. The backup file can subsequently be written onto a tape via a system backup.

!!! WARNING !!!

If a total backup of hard drives is implemented on Windows systems as long as the SQL Server 2012 database is running, the database files (SQL Server) **must not** be saved. Should this occur, this may result in unforeseen malfunctions of the database and hence HYDRA.

If such a complete backup is requested, HYDRA and the SQL Server Database have to be terminated first.

Automatic database backup with HYD-BAM

With **HYD-BAM**, MPDV offers the implementation of an automated backup by means of various maintenance plans.

If the backup is implemented directly onto a **tape drive**, the **system administrator** must only ensure that a new tape is always inserted into the drive and the SQL Server Agent (service) is started. If the backup is implemented onto a local hard drive, sufficient memory capacity must be available in order to accommodate for both the data and the transaction log backups.

The time at which the backup is performed can be set individually in the maintenance plans.

As regards the implementation of an automated backup, 3 and/or 4 **maintenance plans** are set up in the Microsoft SQL Server Management Studio.

1. Maintenance plan for system database backups

The system databases (MSDB, MASTER, MODEL) only change little, but are indispensable. Since their backup needs only a little time, these databases are backed up every day.

2. Maintenance plan for HYDRA database backups

The HYDRA database is backed up completely every day. In backup method 2, the transaction log is also backed up.

3. Maintenance plan for transaction log backups

For backing up the transaction log, a separate maintenance plan is set up.

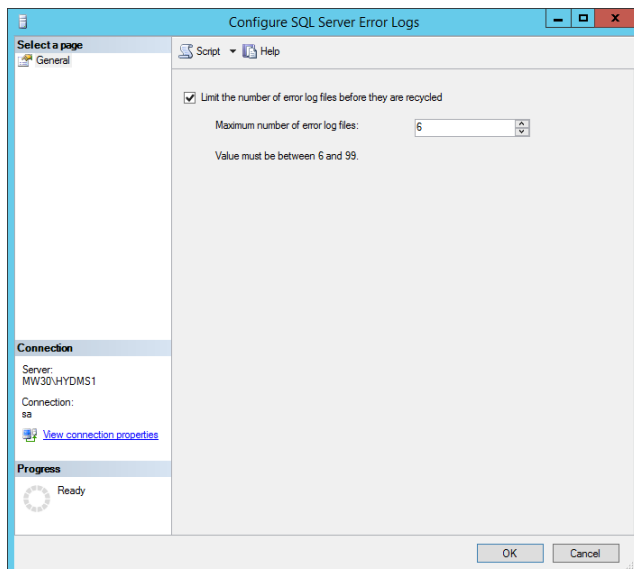
4. Maintenance plan for optimizing the HYDRA database

The HYDRA database is optimized every 7 days.

The time of the HYDRA backup should be between 09:00 PM and 06:00 AM and must be coordinated with the HYDRA deletion scripts.

The logs of the SQL Server 2012 are located in the Management Studio (administration→ SQL Server logs).

Set the maximum number (=6) of SQL Server logs. (administration→SQL Server logs)



HYD-BAM standard settings:

Action	Frequency	Time
Backup of system databases	every day	02:00 AM
Backup of HYDRA database	every day	10:00 PM
Backup of transaction log	hourly	00:00 AM - 12:00 PM
Database optimization	once a week, on Sunday	08:00 AM

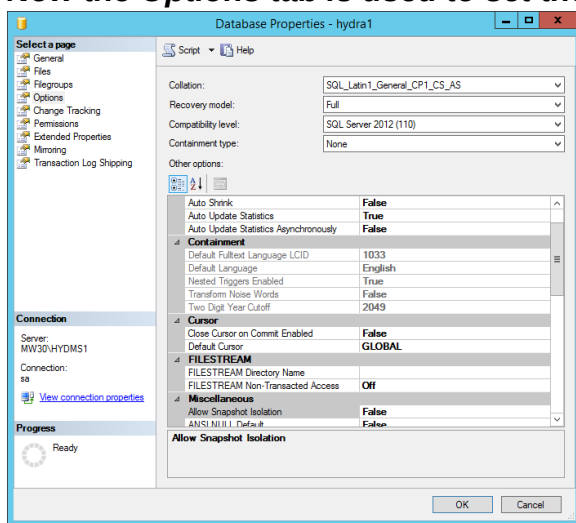
Backup without transaction log backup (method 1)

The backup of the database in method 1 is easy when using the restore mode. Using the model of simple recovery, the database may be restored back to the last backup. However, it is not possible to restore the database back to the time of the error.

The recovery mode is set in:

SERVER NAME → Databases → hydra1 → right mouse button → Properties

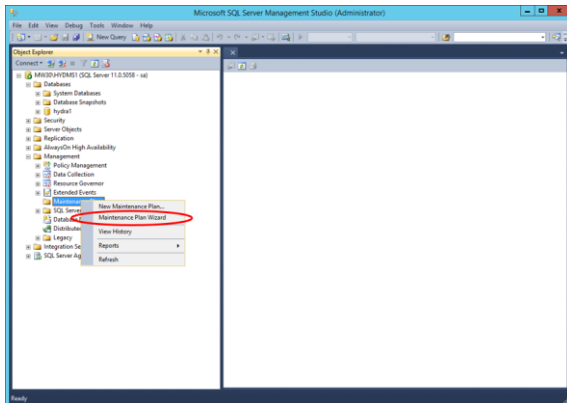
Now the Options tab is used to set the recovery model.



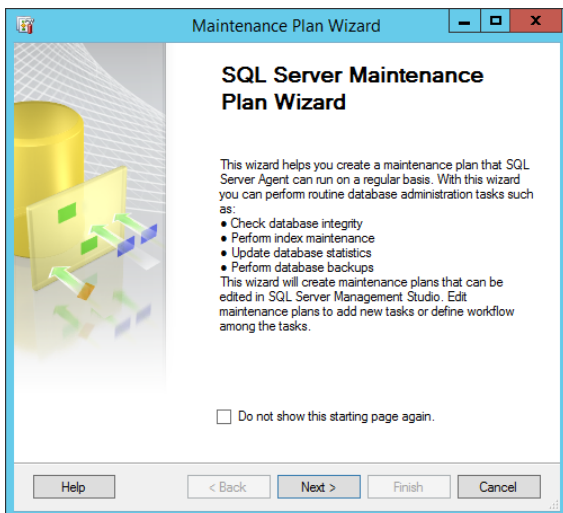
Creation of a maintenance plan using the maintenance plan assistant

- Extend the SQL server instance (server name\HYDMS1) needing a backup in the Object Explorer, and then extend **Administration**.
- Make a right mouse button click on **Maintenance plans**, and then click on **Maintenance plan assistant**

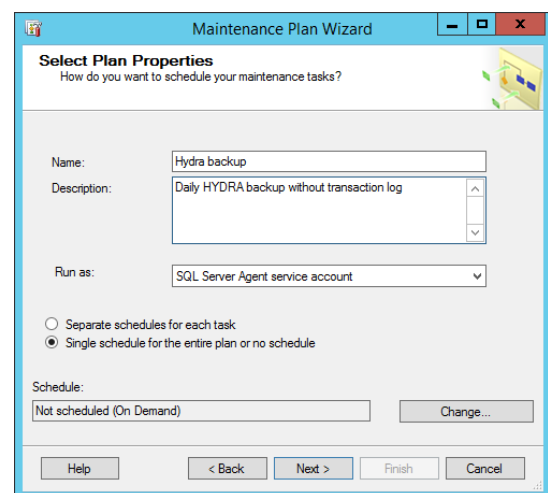
SERVERNAME → Administration → Maintenance plans → Maintenance Plan Wizard



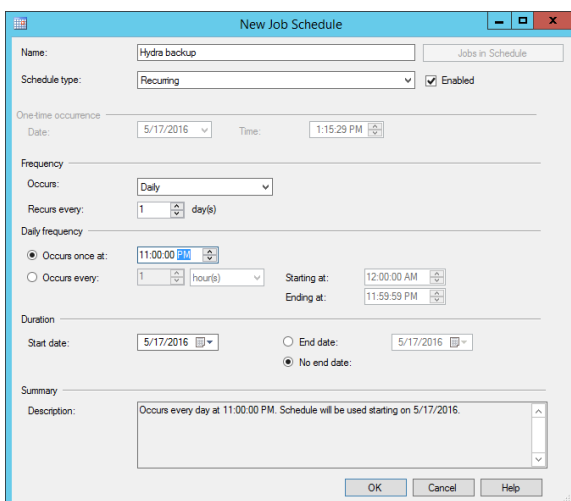
Open Maintenance plan assistant



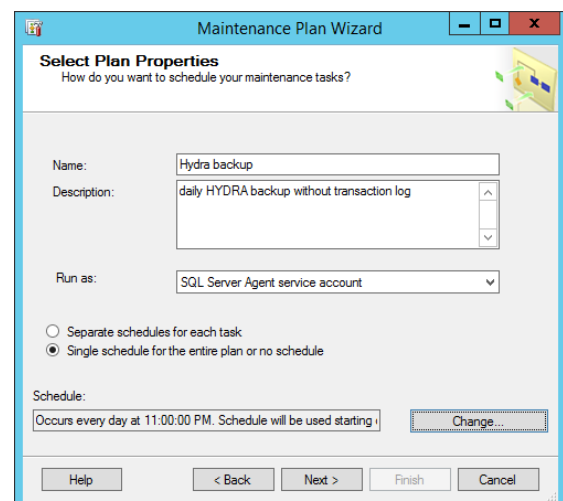
Next



Next

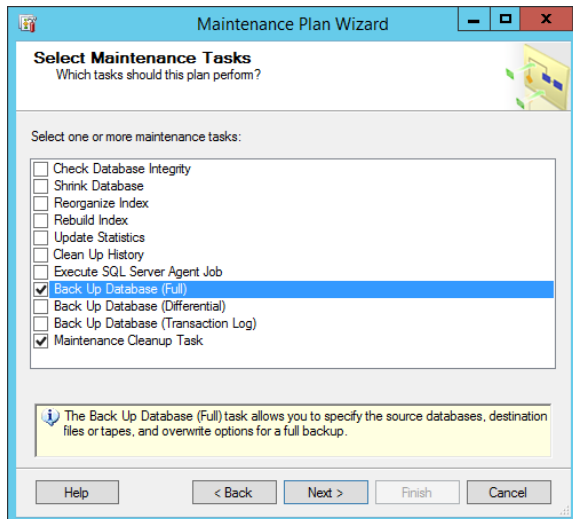


Set time → OK



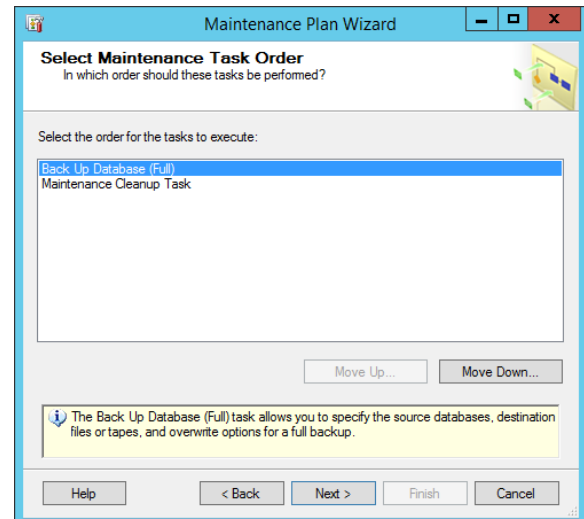
Next

Select the option **Repeated** as **Schedule type** and **daily** for occurrence. The database is to be backed up once at a freely definable time (e.g. 10:00 PM). Click OK to return to the previous screen.

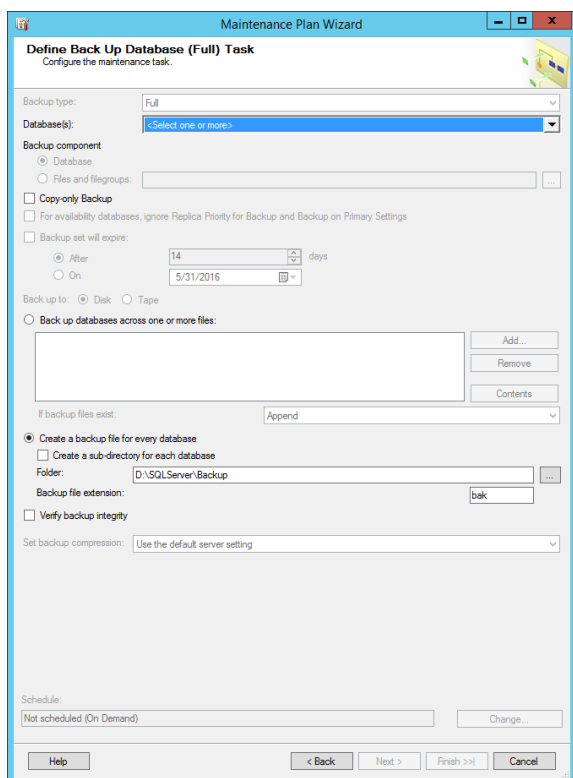


Select tasks: Database backup
Task Maintenance cleanup

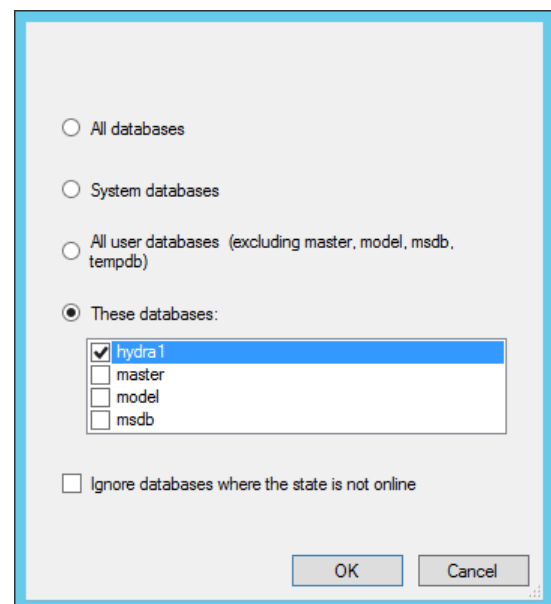
Next



Next



Select database hydra1

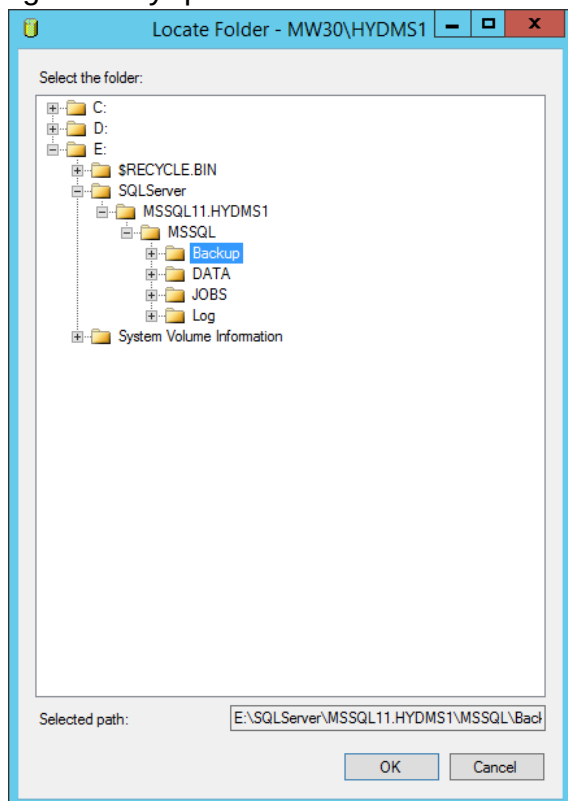


Ok

Select the HYDRA database from specific databases

Backup type:	complete
Backup record expires after:	3 days
Database:	hydra1
Select a backup file for each DB	
Folder:	Indicate backup directory
Backup file extension:	bak

The backup file should be saved on a separate local hard drive with sufficient free memory space. From there, the backup file is subsequently copied to a tape or a backup drive (e.g. 1 hour after the backup) in another step. The database backup should not be made on a tape directly, since the backup on a local hard drive is significantly quicker.

**Please note:**

It may be of benefit to save more than one backup record on the local hard drive. In this case, you select the option "**Backup record expires after**" and set the number of backups to remain on the hard drive (e.g. 3 days). It is to be observed that sufficient hard drive capacity is available.

**Select folder, set number of days to
3 → Next**

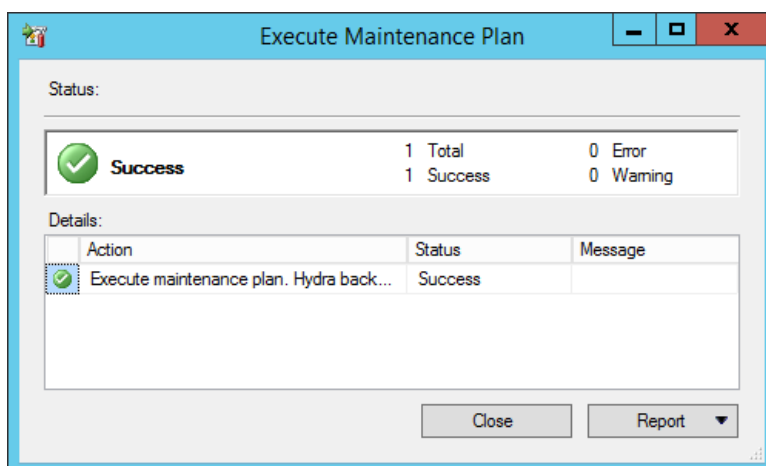
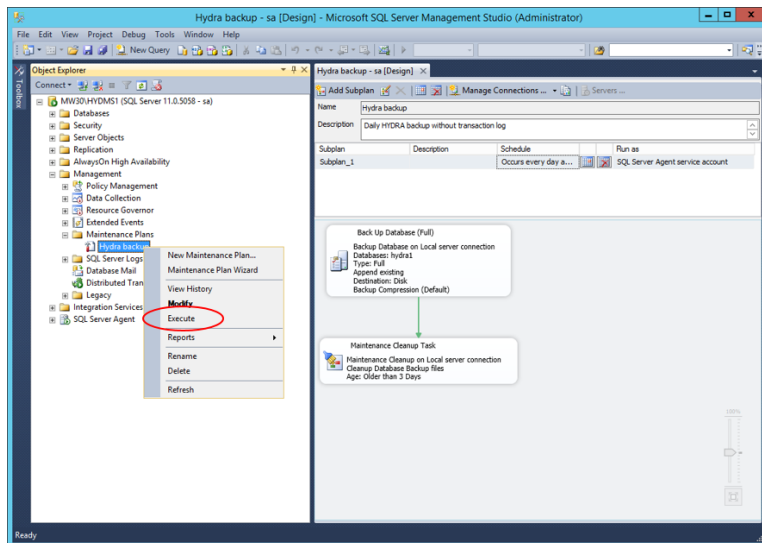
Next

Finish

Action	Status	Message
Creating maintenance plan "Hydra back...	Success	
Adding tasks to the maintenance plan	Success	
Adding scheduling options	Success	
Adding reporting options	Success	
Saving maintenance plan "Hydra backu...	Success	

Close

The new maintenance plan can now be executed and tested.



Close

Check whether the backup file was created in the backup directory. After the daily backup job, the backup must be written onto a tape for additional security.

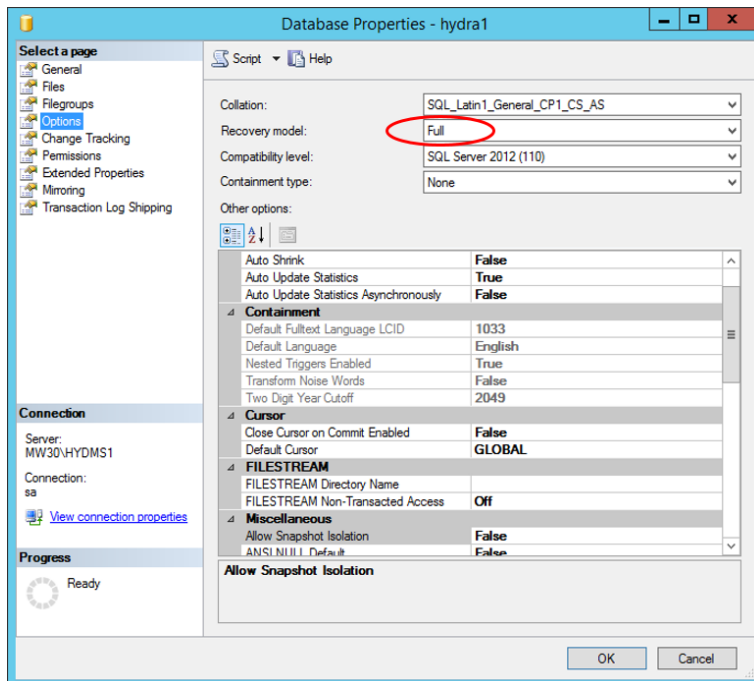
Backup with transaction log backup (method 2)

A prerequisite for method 2 is the use of the complete recovery model. Using the model of complete recovery, the database may be restored back to the time of the error (e.g. hard drive crash).

The recovery mode is set in:

SERVER NAME → Databases → hydra1 → right mouse button → Properties

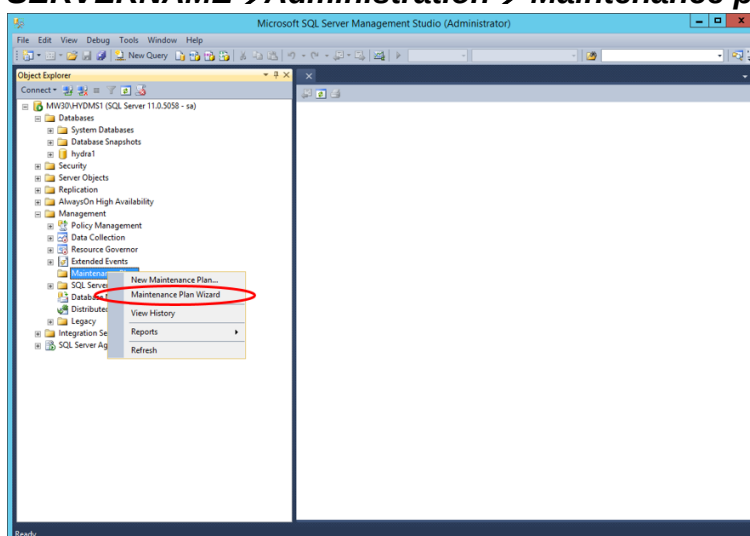
Now the Options tab is used to set the recovery model.

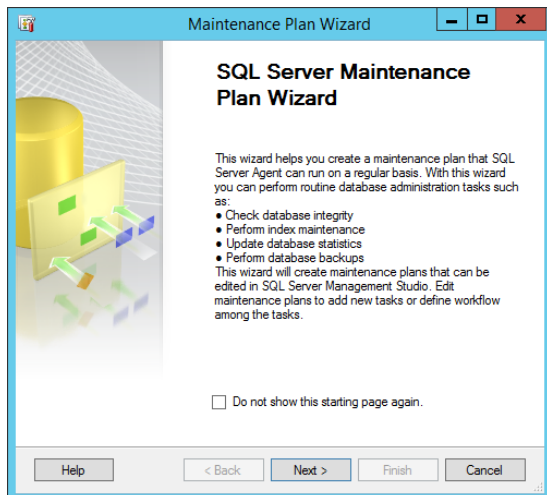


Creation of a maintenance plan using the maintenance plan wizard

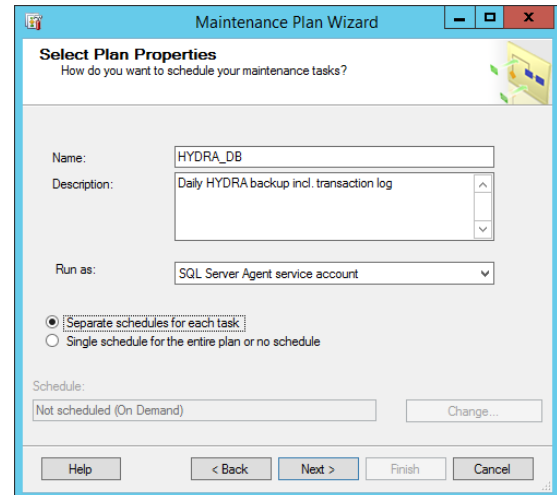
- Extend the SQL server instance (server name\HYDMS1) needing a backup in the Object Explorer, and then extend **Administration**.
- Make a right mouse button click on **Maintenance plans**, and then click on **Maintenance Plan Wizard**

SERVERNAME → Administration → Maintenance plans → Maintenance Plan Wizard



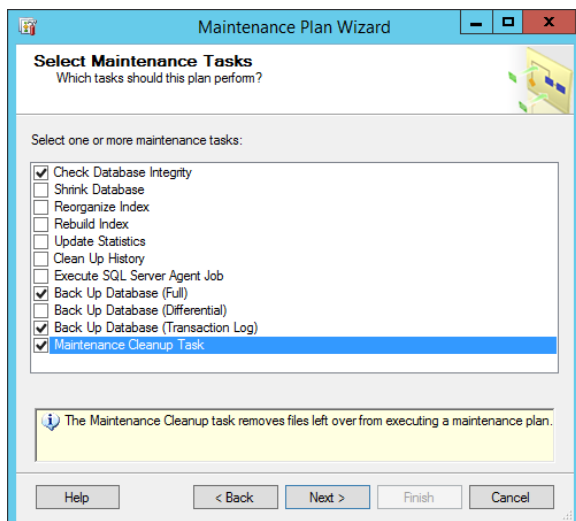


Next



Next

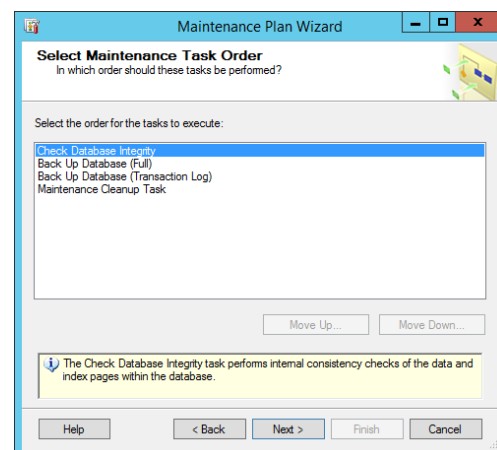
If a common schedule is to be used for all tasks, "One schedule for total" schedule has to be selected here.



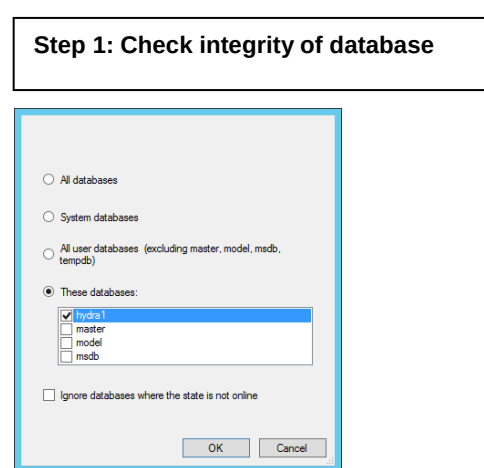
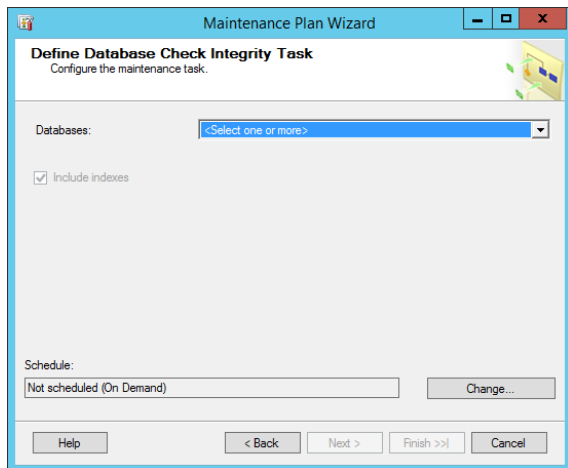
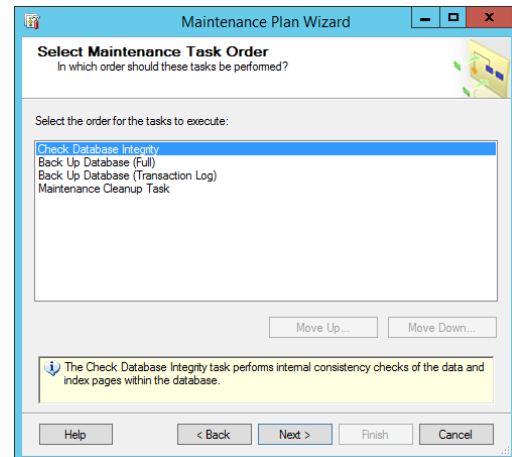
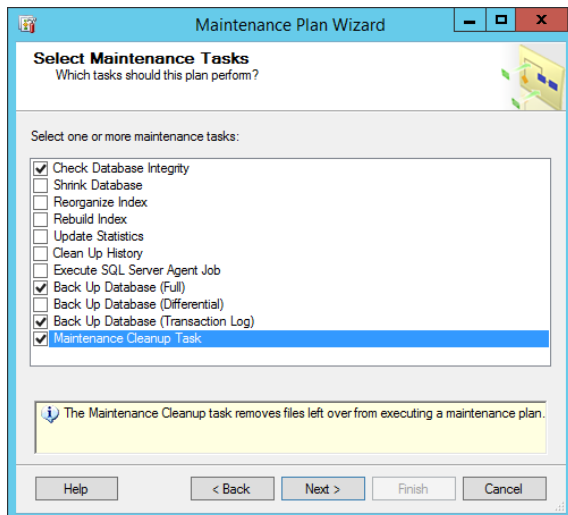
Select maintenance tasks → Next

- Check database integrity
- Back up database (complete)
- Back up database (transaction log)
- Maintenance cleanup task

Next



Next



In this method, all three steps of the backup are performed at three different times.

Step 2: Backup up database

Backup type: full

Backup record expires after:

Database:

Select a backup file for each DB

Folder:

Backup file extension:

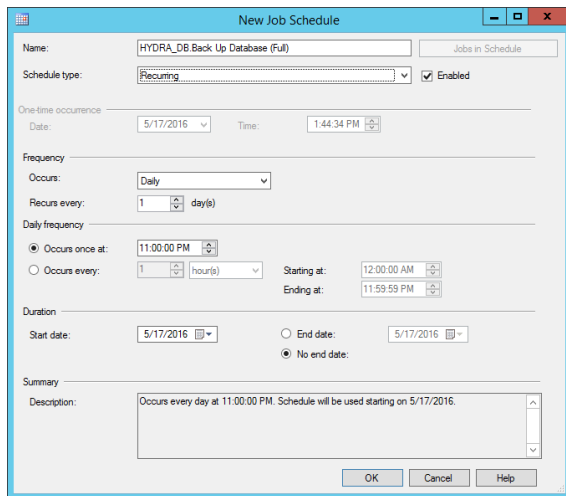
The backup file should be saved on a separate local hard drive with sufficient free memory space. From there, the backup file is subsequently copied to a tape or a backup drive (e.g. 1 hour after the backup) in another step. The database backup should not be made on a tape directly, since the backup on a local hard drive has significant advantages.

3 days

hydra1

Indicate backup directory

bak



New Job Schedule

Name: HYDRA_DB Back Up Database (Full) Jobs in Schedule

Schedule type: Recurring ☒ Enabled

One-time occurrence
Date: 5/17/2016 Time: 1:44:34 PM

Frequency
Occurs: Daily
Recurs every: 1 day(s)

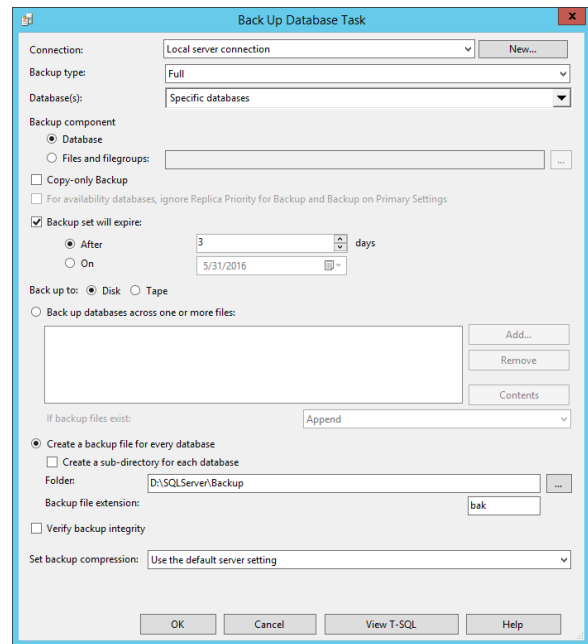
Daily frequency
☒ Occurs once at: 11:00:00 PM
☐ Occurs every: 1 hour(s) Starting at: 12:00:00 AM Ending at: 11:59:59 PM

Duration
Start date: 5/17/2016 ☐ End date: 5/17/2016
☒ No end date:

Summary
Description: Occurs every day at 11:00:00 PM. Schedule will be used starting on 5/17/2016.

OK Cancel Help

OK



Back Up Database Task

Connection: Local server connection New...

Backup type: Full

Database(s): Specific databases

Backup component
☒ Database
☐ Files and filegroups:

☐ Copy-only Backup
For availability databases, ignore Replica Priority for Backup and Backup on Primary Settings

☒ Backup set will expire:
☒ After 3 days
☐ On 5/31/2016

Back up to: ☒ Disk ☐ Tape

☐ Back up databases across one or more files:
Add... Remove Contents

If backup files exist: Append

☒ Create a backup file for every database
☐ Create a sub-directory for each database
Folder: D:\SQLServer\Backup
Backup file extension: bak

☐ Verify backup integrity

Set backup compression: Use the default server setting

OK Cancel View T-SQL Help

OK

Please note:

It may be of benefit to save more than one backup record on the local hard drive. This ensures that you can react very quickly in case of an error. In this case, you select the option **"Backup record expires after"** and set the number of backups to remain on the hard drive (e.g. 3 days). It is to be observed that sufficient hard drive capacity is available.

Step 3: Backup of transaction log

Select the HYDRA database from specific databases

Backup type:

Transaction log

Backup record expires after:

3 days

Database:

hydra1

Select a backup file for each DB

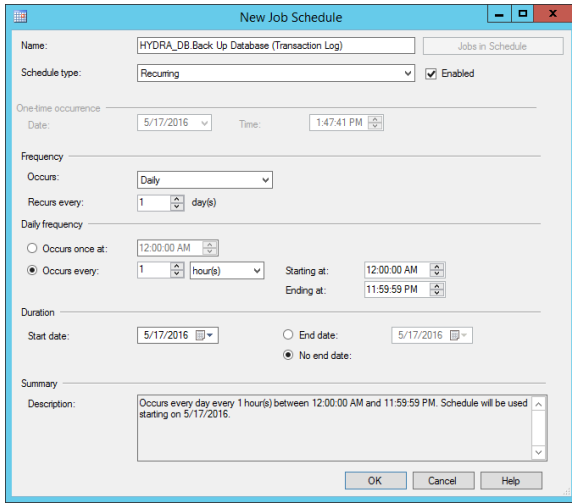
Folder:

Indicate backup directory

Backup file extension:

trn

The backup file should be saved on a separate local hard drive with sufficient free memory space. From there, the backup file is subsequently copied to a tape or a backup drive (e.g. 1 hour after the backup) in another step. The database backup should not be made on a tape directly, since the backup on a local hard drive has significant advantages.



New Job Schedule

Name: HYDRA_DB Back Up Database (Transaction Log) Jobs in Schedule

Schedule type: Recurring ☒ Enabled

One-time occurrence
Date: 5/17/2016 Time: 1:47:41 PM

Frequency
Occurs: Daily
Recurs every: 1 day(s)

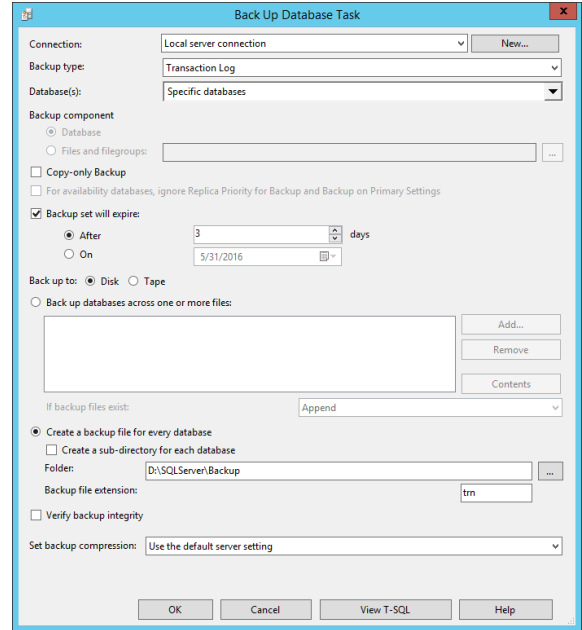
Daily frequency
☐ Occurs once at: 12:00:00 AM
☒ Occurs every: 1 hour(s) Starting at: 12:00:00 AM Ending at: 11:59:59 PM

Duration
Start date: 5/17/2016 ☐ End date: 5/17/2016
☒ No end date

Summary
Description: Occurs every day every 1 hour(s) between 12:00:00 AM and 11:59:59 PM. Schedule will be used starting on 5/17/2016.

OK Cancel Help

OK



Back Up Database Task

Connection: Local server connection New...

Backup type: Transaction Log

Database(s): Specific databases

Backup component
☒ Database
☐ Files and filegroups: ...

☐ Copy-only Backup
☐ For availability databases, ignore Replica Priority for Backup and Backup on Primary Settings

☒ Backup set will expire:
☒ After 3 days
☐ On 5/31/2016

Back up to: ☒ Disk ☐ Tape
☐ Back up databases across one or more files:
... Add... Remove Contents

If backup files exist: Append

☒ Create a backup file for every database
☐ Create a sub-directory for each database
Folder: D:\SQLServer\Backup ...

Backup file extension: .trn

☐ Verify backup integrity

Set backup compression: Use the default server setting

OK Cancel View T-SQL Help

OK

Please note:

It may be of benefit to save more than one backup record on the local hard drive. This ensures that you can react very quickly in case of an error. In this case, you select the option **"Backup record expires after"** and set the number of backups to remain on the hard drive (e.g. 3 days). It is to be observed that sufficient hard drive capacity is available.

Define Maintenance Cleanup Task
Configure the maintenance task.

Delete files of the following type:

- ☒ Backup files
- ☐ Maintenance Plan text reports

File location:

- ☐ Delete specific file
- ☒ Search folder and delete files based on an extension

File name:

Folder:

File extension:

☐ Include first-level subfolders

File age:

- ☒ Delete files based on the age of the file at task run time

Delete files older than the following:

3 Day(s)

Schedule:

Not scheduled (On Demand)

Step 4: Define maintenance cleanup

Select Report Options
Select options for saving or distributing a report of the maintenance plan actions.

- ☒ Write a report to a text file
- ☐ E-mail report

Folder location:

To:

Select folder, set number of days to 3

Next

→Next

The "*" in the field "File extension" will simultaneously delete backup and log files!

Complete the Wizard
Verify the choices made in the wizard, and then click Finish.

Click Finish to perform the following actions:

- Maintenance Plan Wizard
 - Create Maintenance Plan 'HYDRA_DB'
 - Define Database Check Integrity Task
 - Define Back Up Database (Full) Task
 - Define Back Up Database (Transaction Log) Task
 - Define Maintenance Cleanup Task
 - Selected reporting options

Finish

Maintenance Plan Wizard Progress
Click Stop to interrupt the operation.

☒ **Success** 5 Total 0 Error
5 Success 0 Warning

Details:

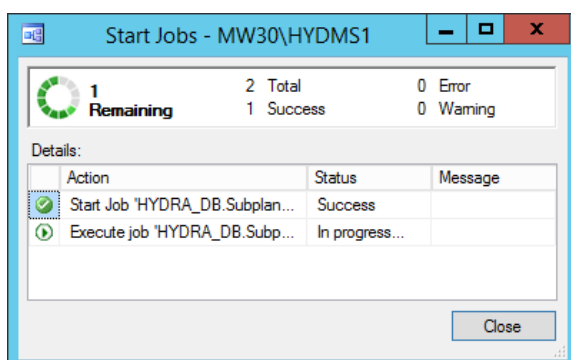
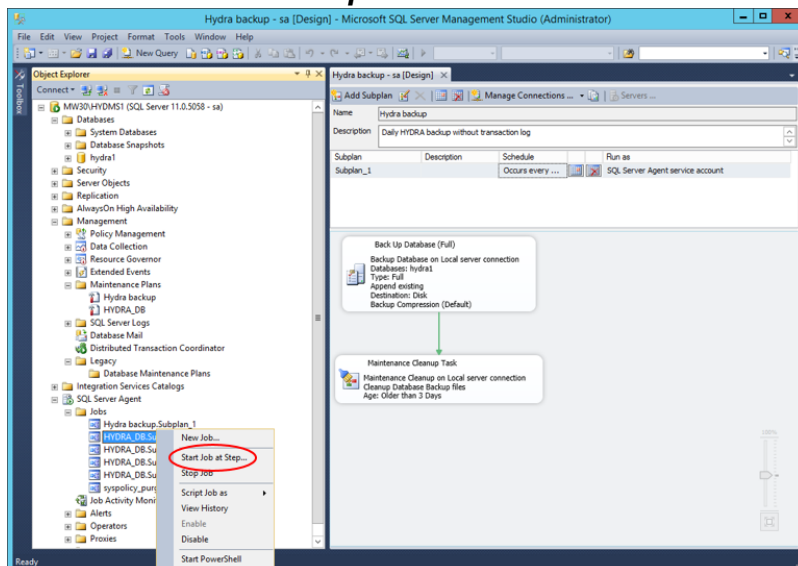
Action	Status	Message
Creating maintenance plan "HYDRA_DB"	Success	
Adding tasks to the maintenance plan	Success	
Adding scheduling options	Success	
Adding reporting options	Success	
Saving maintenance plan "HYDRA_DB"	Success	

Close

The report option allows for saving the log file of the backup in a specific directory. If a mail server has been defined, which is possible in SQL Server Management Studio directly, you may have forwarded the backup report by e-mail or SMS directly if it was successful and/or failed.

In the fourth step, backup files no longer required are now deleted (in this case files older than 3 days) from the backup directory. The administrator is responsible for the smooth implementation of daily deletions. If the backup cannot be implemented because the backup drive is full, the database will stop. As regards the file extension, either a specific extension or "*" for all has to be indicated. If the field remains empty, nothing will be deleted.

The new maintenance plan can now be executed and tested.



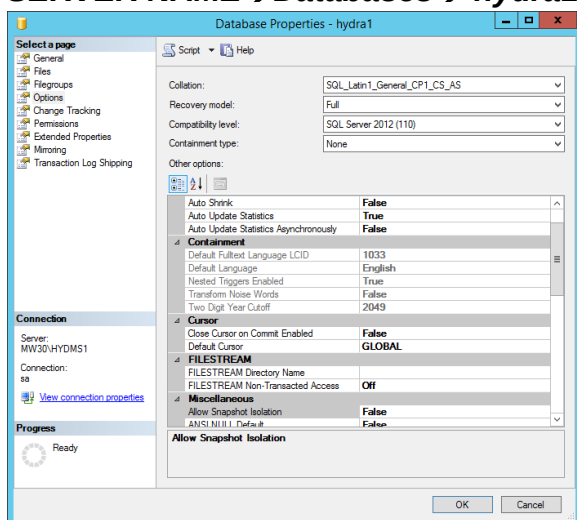
Close

After setting up the maintenance plan, the correct execution of the defined tasks has to be verified daily.

Maintenance plan for system database backups

In the example described the system databases (Master, Model and MSDB) are backed up; the recovery model of the databases Master, Model and MSDB can be set to any mode. In the given case, the databases are set to the simple recovery model. Using the model of simple recovery, the database may be restored back to the last backup. This method is sufficient, since the system databases do not change continuously.

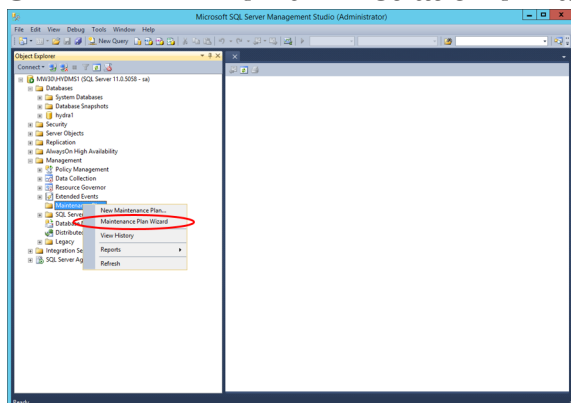
SERVER NAME → Databases → hydra1 → right mouse button → Properties

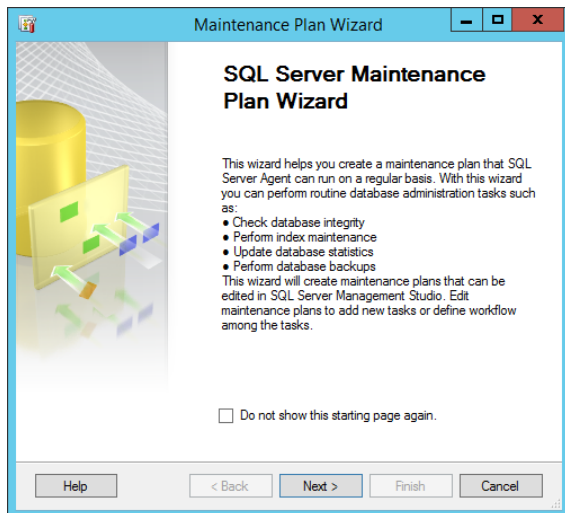


Creation of a maintenance plan using the maintenance plan wizard

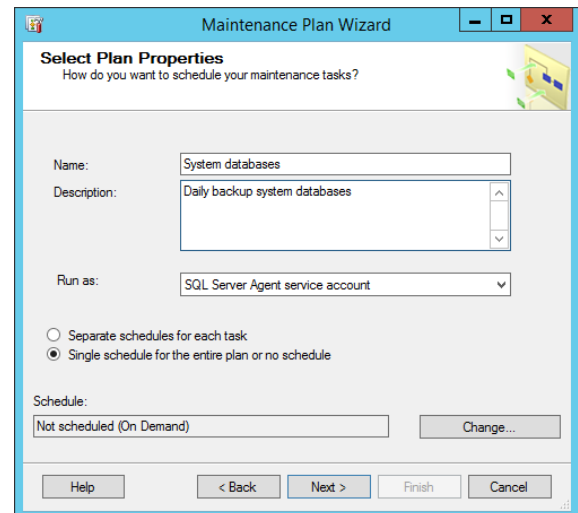
- Extend the SQL server instance (server name\HYDMS1) needing a backup in the Object Explorer, and then extend **Administration**.
- Make a right mouse button click on **Maintenance plans**, and then click on **Maintenance Plan Wizard**

SERVERNAME → Administration → Maintenance plans → Maintenance Plan Wizard

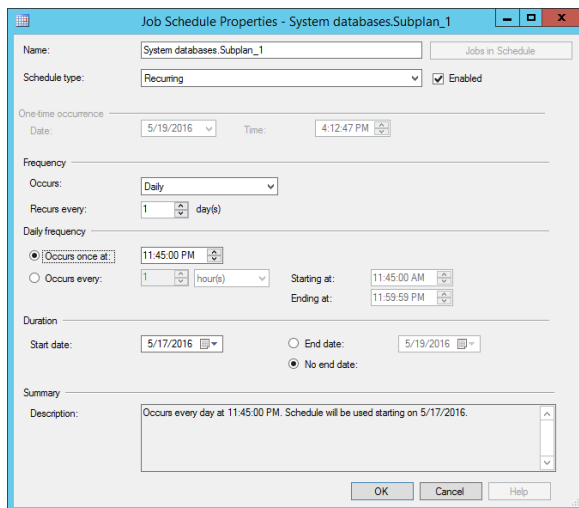




Next

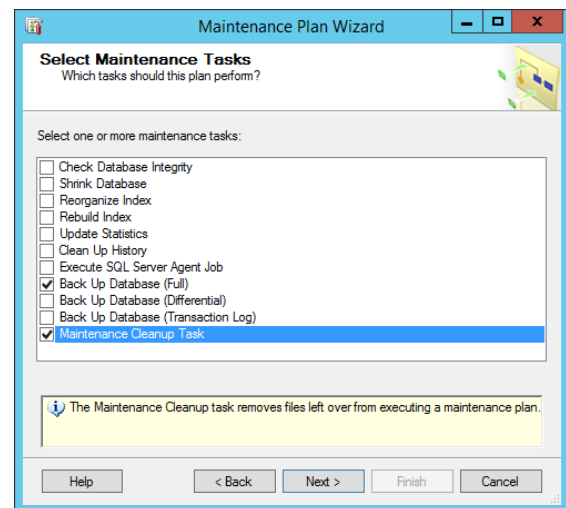


Change

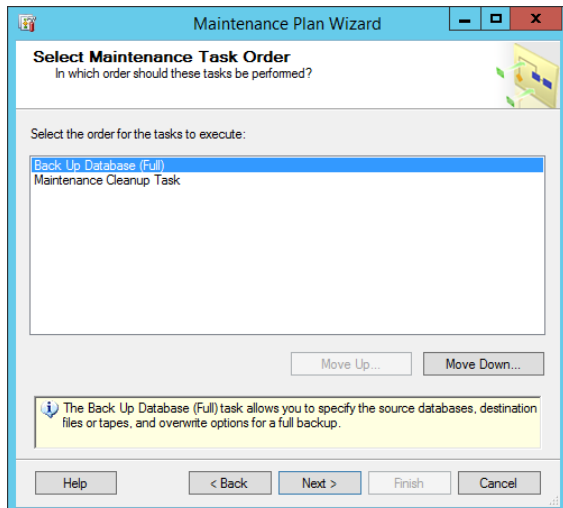


Time type: **repeated**

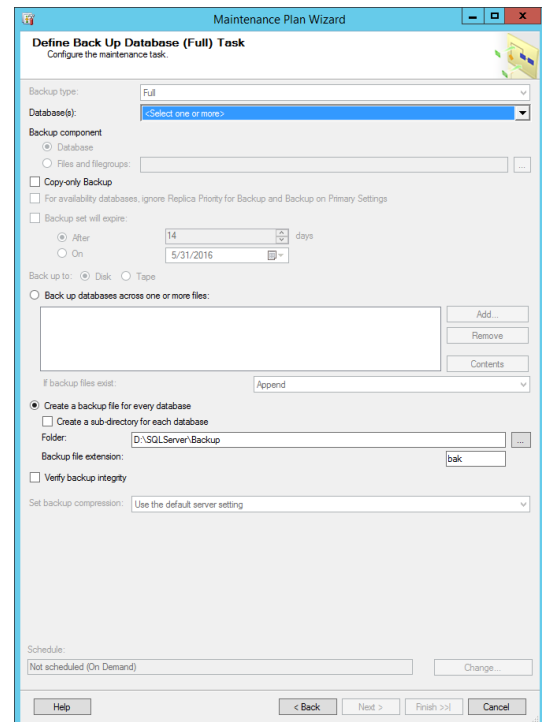
Occurrence: **daily**
Once at: **e.g. 11:45 PM**
No end date
OK



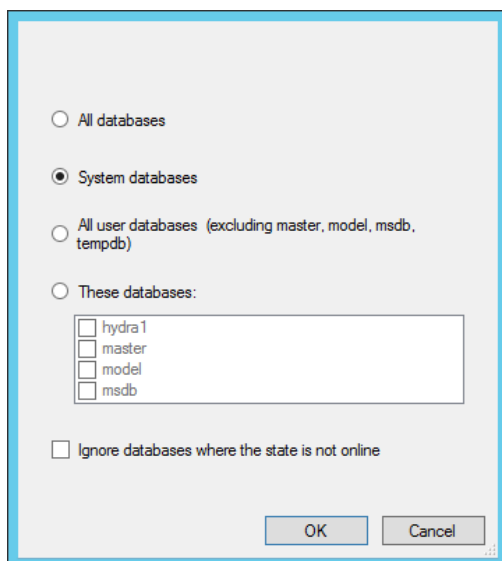
Step 1: Backup of system databases



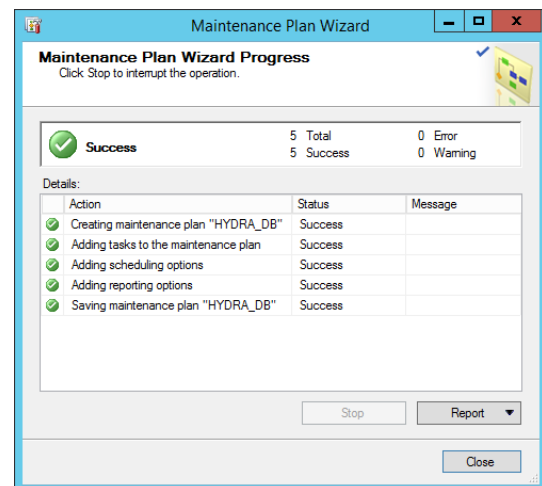
Select database backup
Select Maintenance cleanup
Next



Next



Backup type:
Backup record expires after:
Database:
Select a backup file for each DB
Folder:
Backup file extension:



complete
3 days
System databases

Indicate backup directory
bak

The backup file should be saved on a separate local hard drive with sufficient free memory space. From there, the backup file is subsequently copied to a tape or a backup drive (e.g. 1 hour after the backup) in another step. The database backup should not be made on a tape directly, since the backup on a local hard drive has significant advantages.

Define Maintenance Cleanup Task
Configure the maintenance task.

Delete files of the following type:

- ☒ Backup files
- ☐ Maintenance Plan text reports

File location:

- ☐ Delete specific file

File name:
- ☒ Search folder and delete files based on an extension

Folder:

File extension:

☐ Include first-level subfolders

File age:

- ☒ Delete files based on the age of the file at task run time

Delete files older than the following:

3 Day(s)

Schedule:

Step 2:maintenance cleanup

Select Report Options
Select options for saving or distributing a report of the maintenance plan actions.

- ☒ Write a report to a text file

Folder location:
- ☐ Email report

To:

Select folder, set number of months to 3 → Continue

Complete the Wizard
Verify the choices made in the wizard, and then click Finish.

Click Finish to perform the following actions:

- Maintenance Plan Wizard
 - + Create Maintenance Plan 'System databases'
 - + Define Back Up Database (Full) Task
 - + Define Maintenance Cleanup Task
 - + Selected reporting options

Finish

Maintenance Plan Wizard Progress
Click Stop to interrupt the operation.

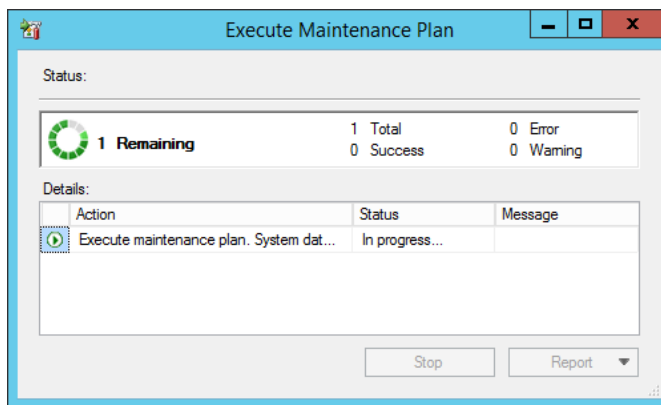
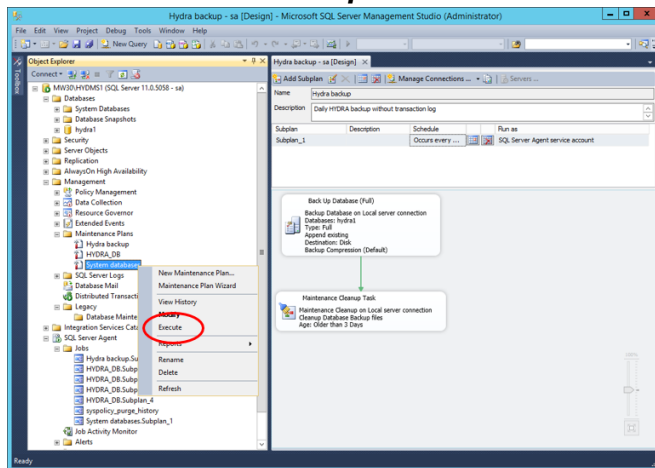
☒ **Success** 5 Total 0 Error
5 Success 0 Warning

Details:

Action	Status	Message
Creating maintenance plan "System dat...	Success	
Adding tasks to the maintenance plan	Success	
Adding scheduling options	Success	
Adding reporting options	Success	
Saving maintenance plan "System data...	Success	

Close

The new maintenance plan can now be executed and tested.



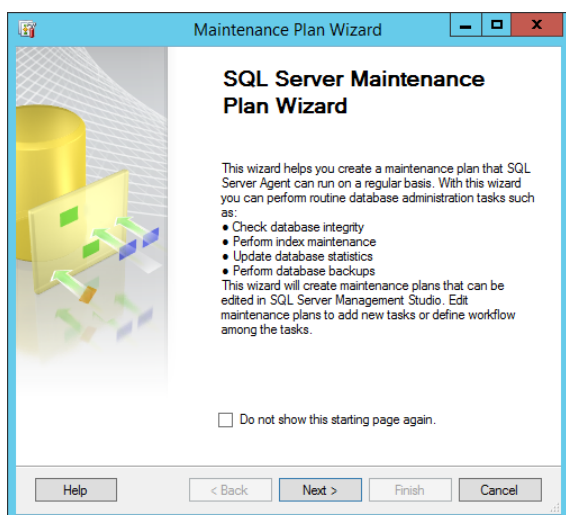
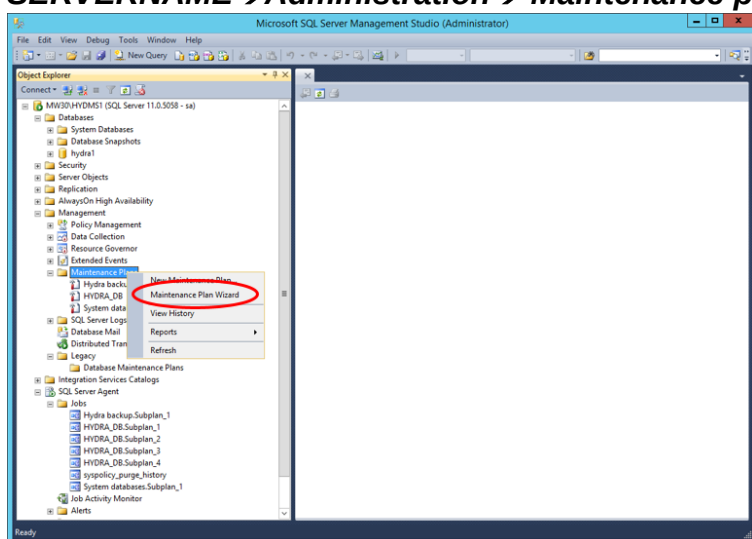
After setting up the maintenance plan, the correct execution of the defined tasks has to be verified daily.

Maintenance plan to optimize the HYDRA database

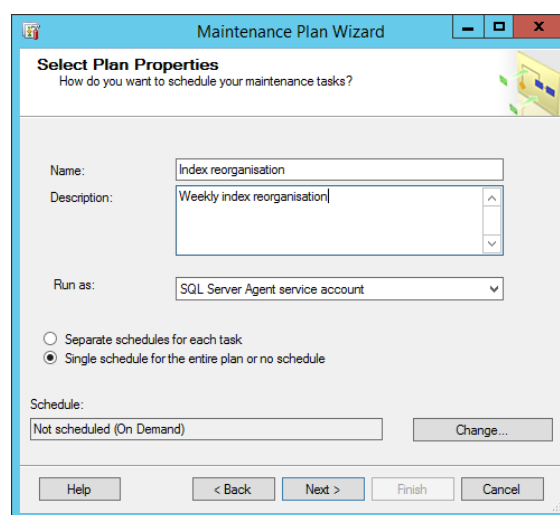
You proceed as follows in order to set up a maintenance plan for optimizing the HYDRA database:

- Open SQL Server Management Studio
- Extend the SQL server instance (server name\HYDMS1) needing a backup in the Object Explorer, and then extend **Administration**.
- Make a right mouse button click on Maintenance plans, and then click on Maintenance Plan Wizard

SERVERNAME → Administration → Maintenance plans → Maintenance Plan Wizard



Next



Change

New Job Schedule

Name: Jobs in Schedule

Schedule type: ☒ Enabled

One-time occurrence
Date: Time:

Frequency
Occurs:
Recur every: week(s) on
☐ Monday ☐ Wednesday ☐ Friday ☐ Saturday
☐ Tuesday ☐ Thursday ☒ Sunday

Daily frequency
☒ Occurs once at:
☐ Occurs every: hour(s) Starting at: Ending at:

Duration
Start date: ☐ End date:
☒ No end date

Summary
Description:

OK Cancel Help

Time type: *repeated*
Occurrence: *weekly*
Repeat every: *Sunday*
Once at: *e.g. 08:00 AM*
No end date

Maintenance Plan Wizard

Select Maintenance Tasks
Which tasks should this plan perform?

Select one or more maintenance tasks:

- ☒ Check Database Integrity
- ☐ Shrink Database
- ☒ Reorganize Index
- ☐ Rebuild Index
- ☐ Update Statistics
- ☐ Clean Up History
- ☐ Execute SQL Server Agent Job
- ☐ Back Up Database (Full)
- ☐ Back Up Database (Differential)
- ☐ Back Up Database (Transaction Log)
- ☐ Maintenance Cleanup Task

The Reorganize Index task defragments and compacts clustered and non-clustered indexes on tables and views. This will improve index-scanning performance.

Help < Back Next > Finish Cancel

**" Check database integrity and
Select "Reorganize index"
Continue**

Maintenance Plan Wizard

Select Maintenance Task Order
In which order should these tasks be performed?

Select the order for the tasks to execute:

- Check Database Integrity
- Reorganize Index

Move Up... Move Down...

The Check Database Integrity task performs internal consistency checks of the data and index pages within the database.

Help < Back Next > Finish Cancel

Next

Maintenance Plan Wizard

Define Database Check Integrity Task
Configure the maintenance task.

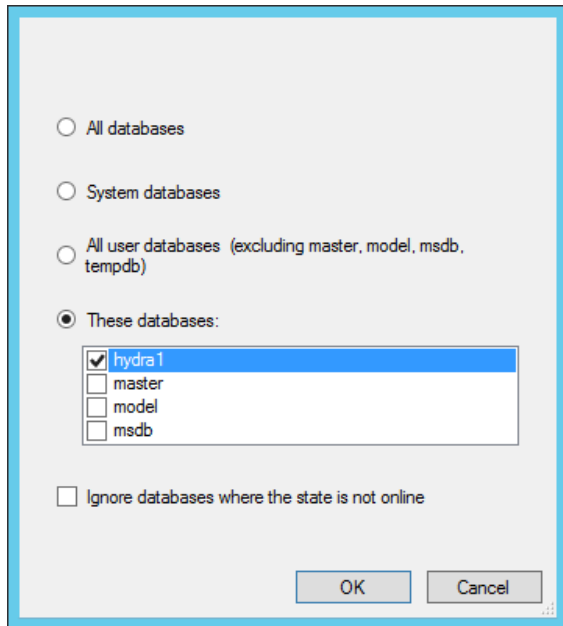
Databases:

☒ Include indexes

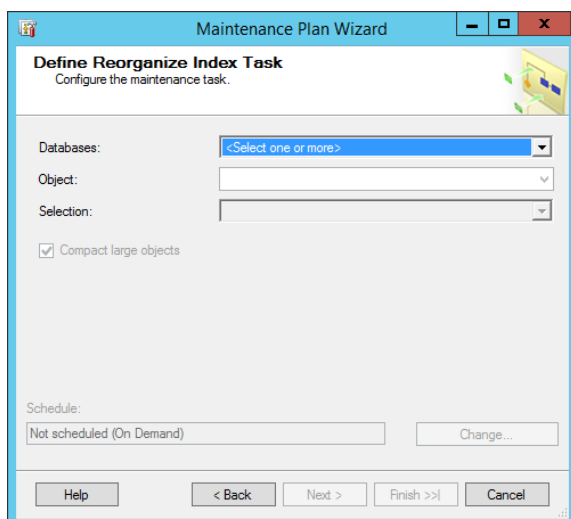
Schedule: Change...

Help < Back Next > Finish >> Cancel

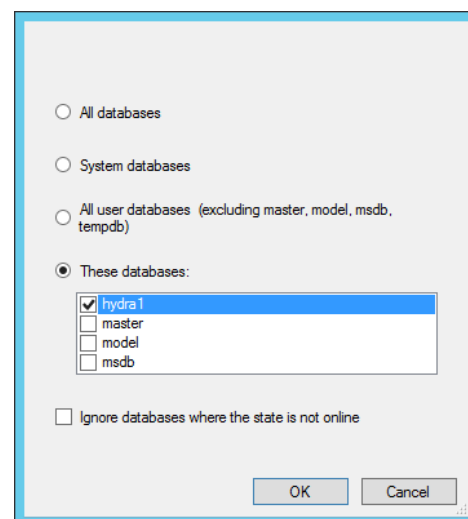
Select database



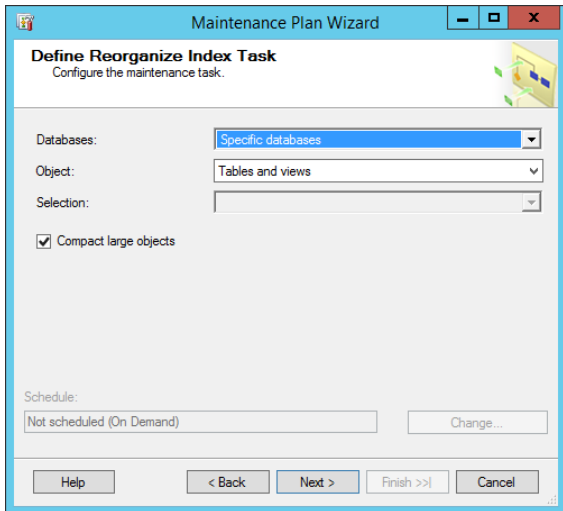
Select HYDRA1 database →OK



Select database



Select HYDRA1 database →OK



Maintenance Plan Wizard

Define Reorganize Index Task
Configure the maintenance task.

Databases:

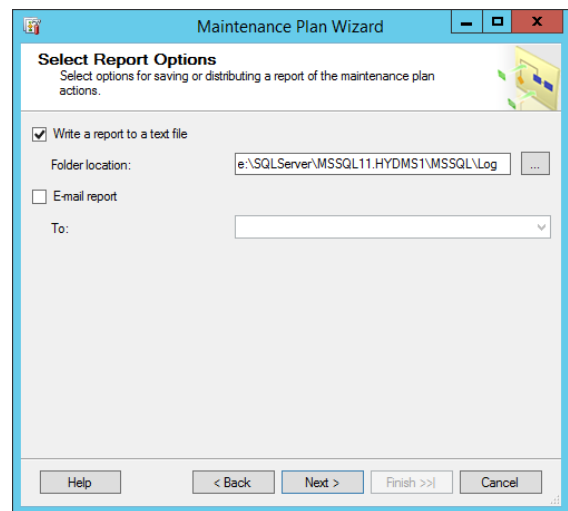
Object:

Selection:

☒ Compact large objects

Schedule:

Next



Maintenance Plan Wizard

Select Report Options
Select options for saving or distributing a report of the maintenance plan actions.

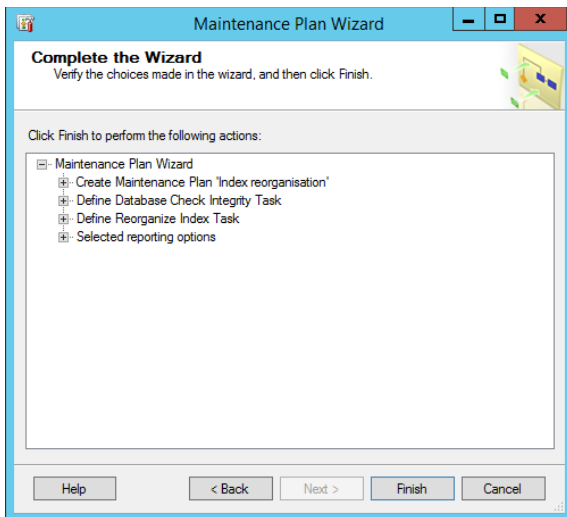
☒ Write a report to a text file

Folder location:

☐ E-mail report

To:

Next



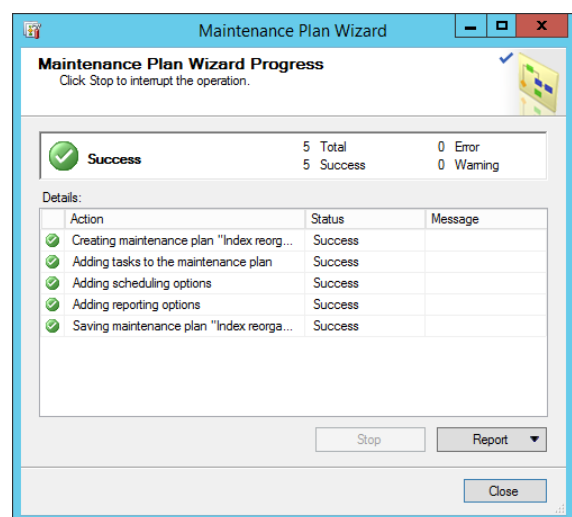
Maintenance Plan Wizard

Complete the Wizard
Verify the choices made in the wizard, and then click Finish.

Click Finish to perform the following actions:

- Maintenance Plan Wizard
 - Create Maintenance Plan "Index reorganisation"
 - Define Database Check Integrity Task
 - Define Reorganize Index Task
 - Selected reporting options

Finish



Maintenance Plan Wizard Progress
Click Stop to interrupt the operation.

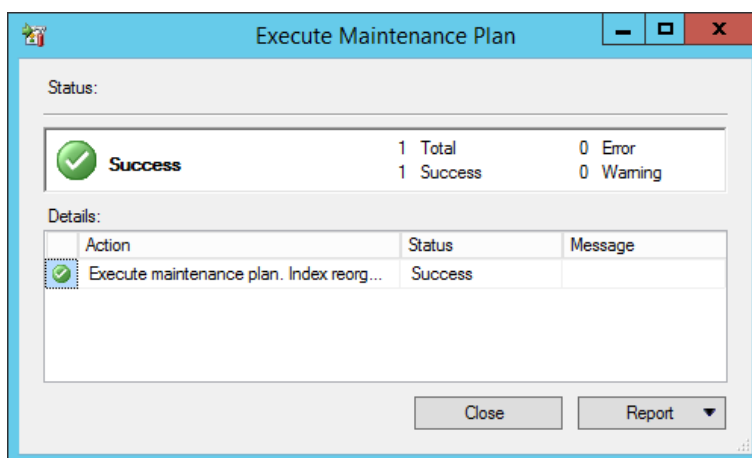
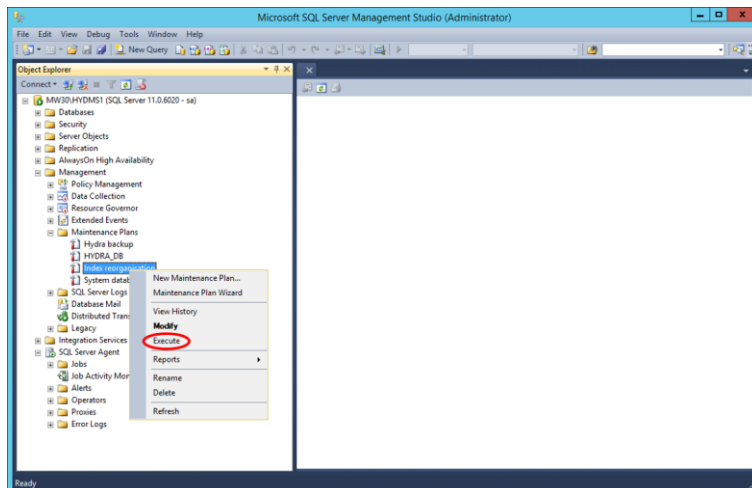
☒ **Success** 5 Total 0 Error
5 Success 0 Warning

Details:

Action	Status	Message
Creating maintenance plan "Index reorg...	Success	
Adding tasks to the maintenance plan	Success	
Adding scheduling options	Success	
Adding reporting options	Success	
Saving maintenance plan "Index reorga...	Success	

Close

Subsequently, you can execute and test the maintenance plan generated.



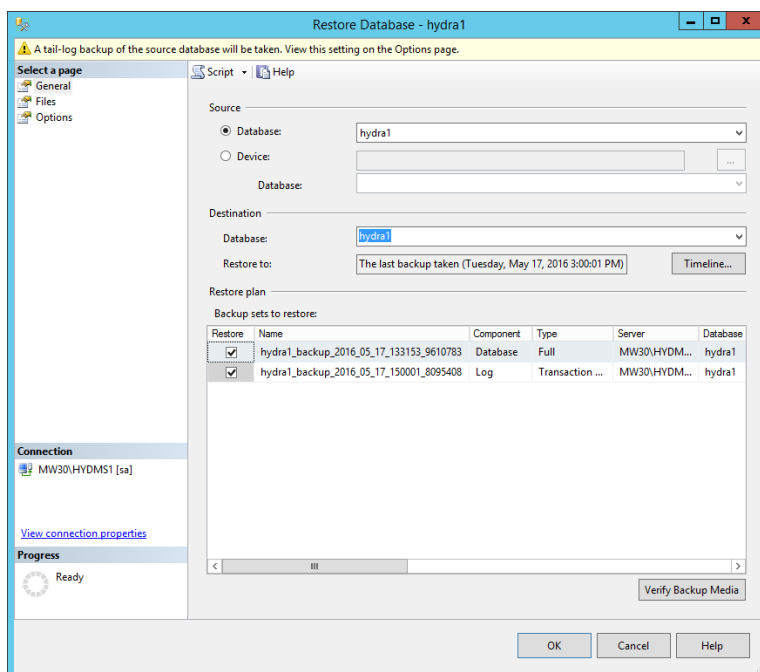
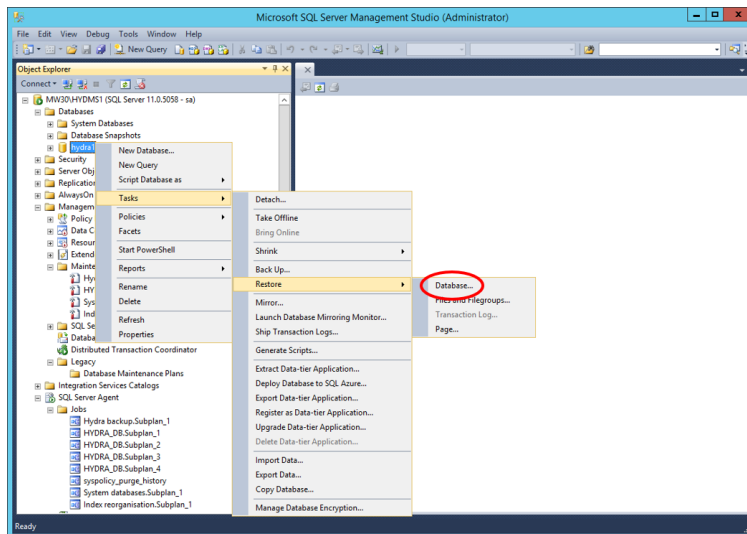
Close

2 Restore data (RESTORE AND RECOVERY)

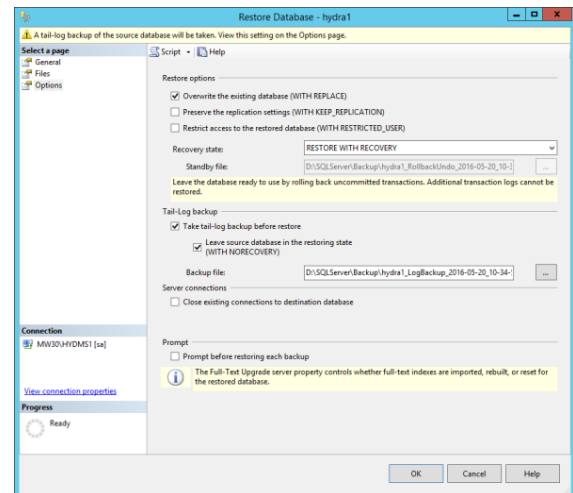
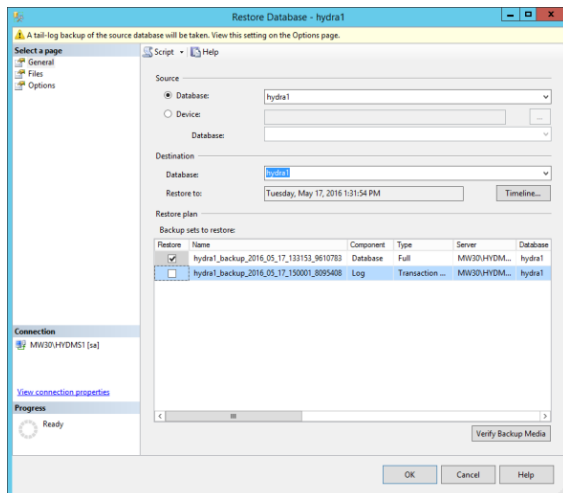
Restore without transaction log backup

Please note: At the time of restore, no user must be connected to the database. All HYDRA services must be terminated so that a database backup may be loaded.

- Open SQL Server Management Studio
- Extend the SQL server instance (server name\HYDMS1) needing a backup in the **Object Explorer**.
- Extend the **Databases** group, make a right mouse button click on the **HYDRA1** database, go to **Tasks** and then to Restore. Now click on **Restore database**.



SQL Server now shows the current backups known to the system. In the given example, these are a **Data backup** and a **Transaction log backup** for the selected **HYDRA1** database. In this example, only the **Database** component of the type is selected completely, the existing transaction log is not loaded.



If the backup record is requested to be restored from another medium (e.g. a hard drive), and the backup record is not recognized by SQL Server as the backup instance automatically, the procedure described in item 4.4 is to be followed.

By means of the **Options** part of the **Restore database** dialog field, you may enter additional options to restore a database.

Overwrite the existing database

This specifies that all existing databases and the related files are overwritten in the restore process, even if a database or file with the same name exists already.

Retain replication settings (not to be used for HYDRA1 database)

This retains the replication settings, if a published database is restored on a server where the database was not created.

Confirmation before recovery of individual backup

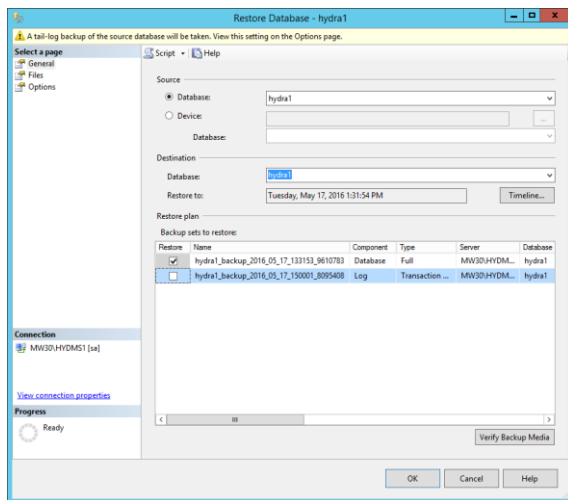
This requires a confirmation from the user with each backup record, before restore is started. This option is particularly helpful if you have to change the tapes for various media records, e.g. if the server has only one tape device.

Restrict access to restored database

→ Restricted mode (no user access possible)

Only allows members of **db_owner**, **dbcreator** or **sysadmin** to access the restored database. This option corresponds to the option **RESTRICTED_USER** in a transact SQL **RESTORE** instruction.

Enter the name of the database to be restored in the field **Target to be restored as database**, if it deviates from the default, or select the name. If you wish to restore the database under a **new name**, you enter the new name of the database.

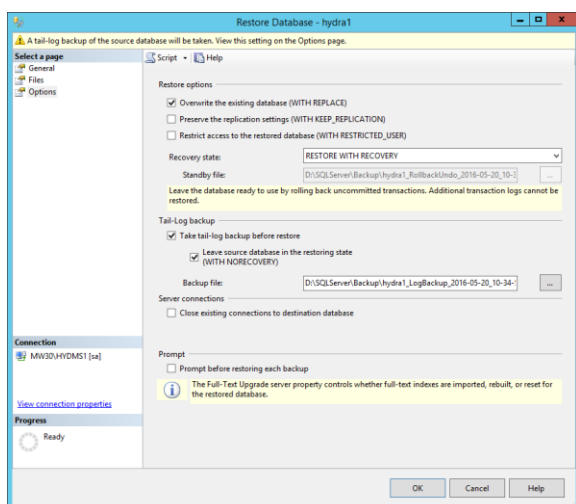


Select the backup record from where you intend to restore the database from **Recovery source**. In this method, the transaction log must not be indicated.

If the backup record is to be taken from **another medium**, this medium has to be selected first. (see section 4.4)

You may optionally click on the **Options** tab and execute the following actions:

Set the recovery option **Overwrite the existing database**, if the database (HYDRA1) still exists.



Leave the database operational by executing a rollback for transactions without commit. Additional transaction logs cannot be restored. (RESTORE WITH RECOVERY)

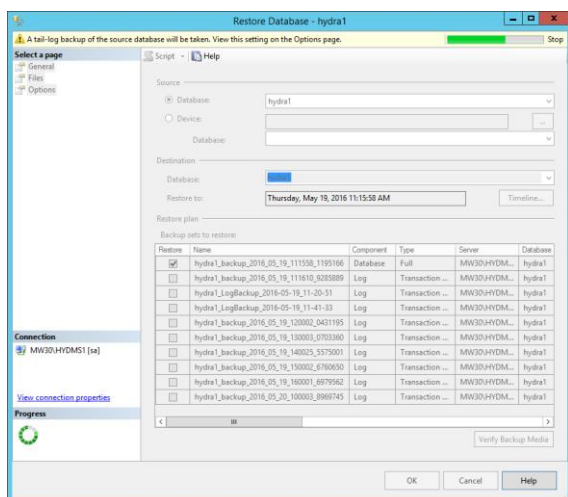
If transaction logs are indicated, these are already loaded with Restore. **If possible, use the first restore method.**

Do not leave the database operational and do not execute a rollback for transactions without commit. Additional transaction logs can be restored. (RESTORE WITH NORECOVERY)

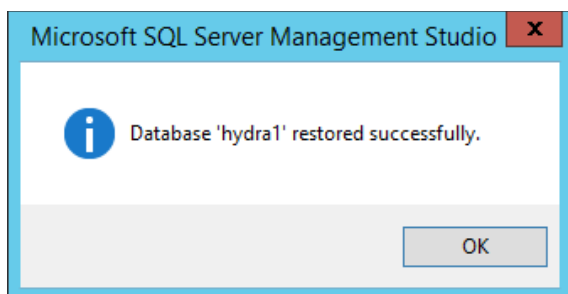
If additional **transaction logs** are available, e.g. the last transaction log created manually. The recovery instruction must then be given manually.

Leave database in write-protected mode. Transactions without commit are reversed, the reversal actions, however, are saved in a standby file so that the effects of recovery may be reversed. (RESTORE WITH STANDBY)

This option is used, if you are not sure whether the uploaded status has to be verified first. The administrator may check the data without restoring the database and without opening it for the users.



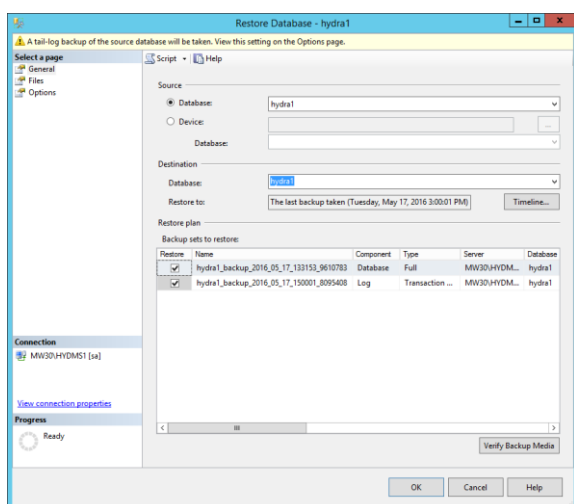
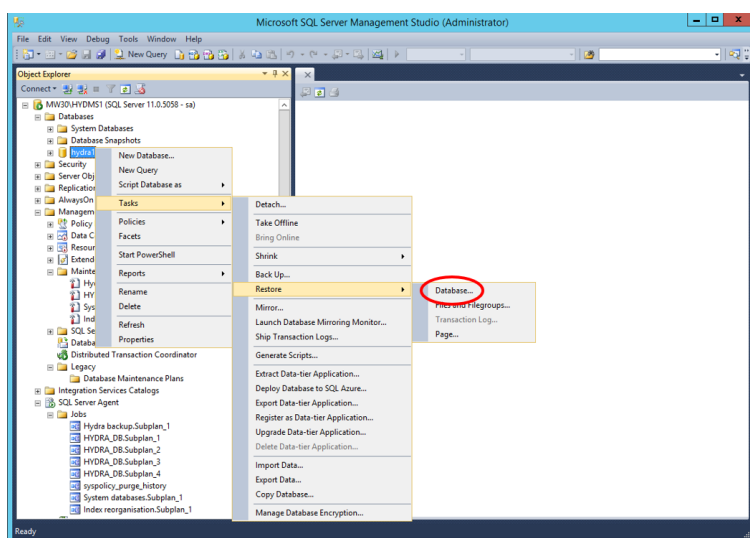
If everything is successful, the database has been restored successfully.



Restore with transaction log backup

Please note: At the time of restore, no user must be connected to the database. All HYDRA services must be terminated so that a database backup may be loaded.

- Open SQL Server Management Studio
- Extend the SQL server instance (server name\HYDMS1) needing a backup in the **Object Explorer**.
- Extend the **Databases** group, make a right mouse button click on the **HYDRA1** database, go to **Tasks** and then to **Restore**. Now click on **Restore database**.



SQL Server now shows the current backups known to the system. In the given example, these are a complete **Data backup** and a **Transaction log backup** for the selected HYDRA1 database. In the example, both components are selected, the existing transaction log is loaded.

In this example, the database recovery is indicated with the option **Latest possible time**. Depending on the error case, another option has to be indicated here (e.g. time-based or date-related).

If the backup record is requested to be restored from another medium (e.g. a hard drive), and the backup record is not recognized by SQL Server as the backup instance automatically, the procedure described in item 4.4 is to be followed.

By means of the **Options** part of the **Restore database** dialog field, you may enter additional options to restore a database.

Overwrite the existing database

This specifies that all existing databases and the related files are overwritten in the restore process, even if a database or file with the same name exists already.

Retain replication settings (not to be used for HYDRA1 database)

This retains the replication settings, if a published database is restored on a server where the database was not created.

Confirmation before recovery of individual backup

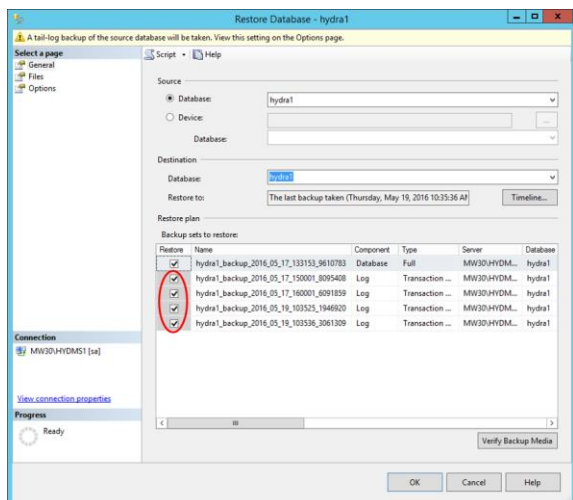
This requires a confirmation from the user with each backup record, before restore is started. This option is particularly helpful if you have to change the tapes for various media records, e.g. if the server has only one tape device.

Restrict access to restored database

→ Restricted mode (no user access possible)

Only allows members of **db_owner**, **dbcreator** or **sysadmin** to access the restored database. This option corresponds to the option **RESTRICTED_USER** in a transact SQL **RESTORE** instruction.

Enter the name of the database to be restored in the field **Target to be restored as database**, if it deviates from the default, or select the name. If you wish to restore the database under a new name, you enter the new name of the database.



Select the backup record from where you intend to restore the database from **Recovery source**. In this method, the transaction log must also be indicated.

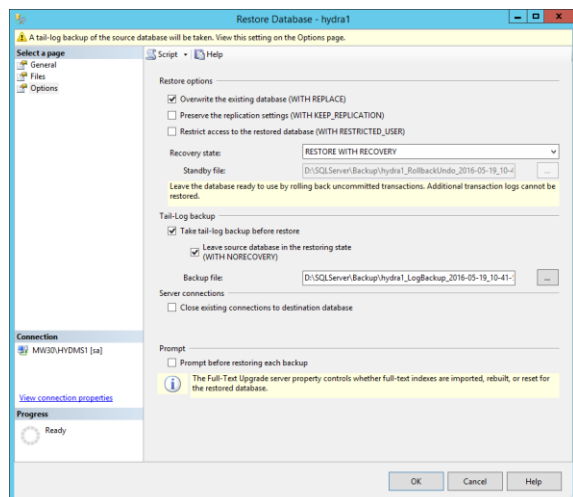
You may optionally click on the **Options** tab and execute the following actions:

Click on **Do not leave the database operational** and do not execute a rollback for transactions without commit. **Additional transaction logs can be restored (RESTORE WITH NORECOVERY)**.

If the backup record is to be taken from **another medium**, this medium has to be selected first. (See section 4.4)

You may optionally click on the **Options** tab and execute the following actions:

Set the recovery option **Overwrite the existing database**, if the database (HYDRA1) still exists.



Leave the database operational by executing a rollback for transactions without commit. Additional transaction logs cannot be restored. (RESTORE WITH RECOVERY)

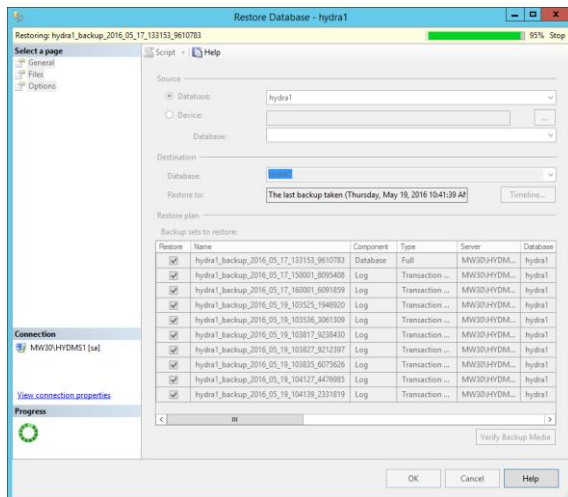
If transaction logs are indicated, these are already loaded with Restore. **If possible, use the first restore method.**

Do not leave the database operational and do not execute a rollback for transactions without commit. Additional transaction logs can be restored. (RESTORE WITH NORECOVERY)

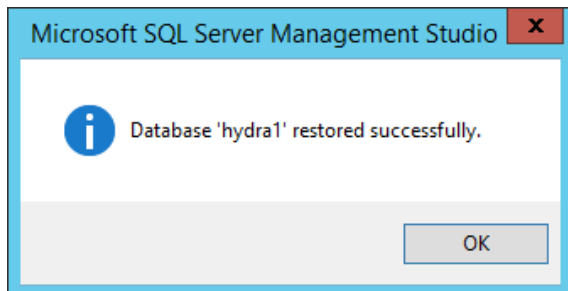
If other transaction logs are available (e.g. the latest transaction log created manually), the recovery instruction must be given manually.

Leave database in write-protected mode. Transactions without commit are reversed, the reversal actions, however, are saved in a standby file so that the effects of recovery may be reversed. (RESTORE WITH STANDBY)

This option is used, if you are not sure whether the uploaded status has to be verified first. The administrator may check the data without restoring the database or opening it for the users.

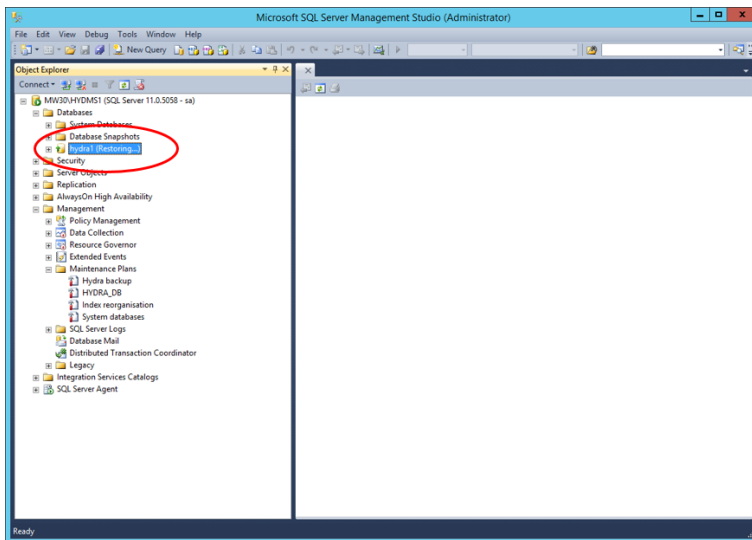


If everything is successful, the database has been restored successfully.



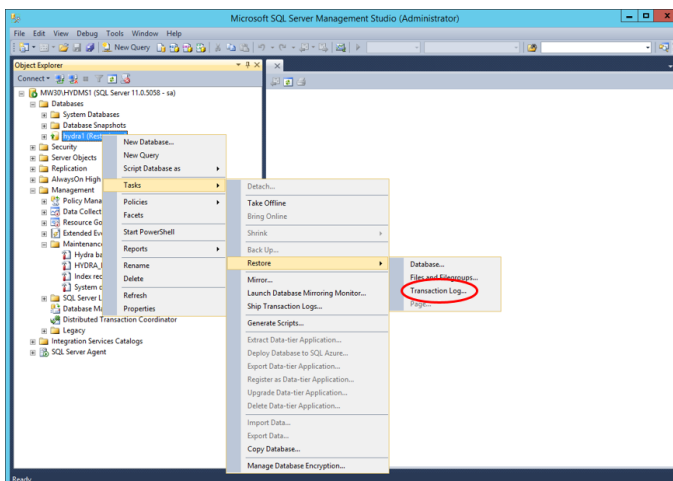
Loading a transaction log after restore by using the option RESTORE WITH NORECOVERY

After the database has been restored (RESTORE WITH NORECOVERY), the transaction log is now restored. As shown in the figure below, the database remains in the "restored" status until the transaction log backup is completed.

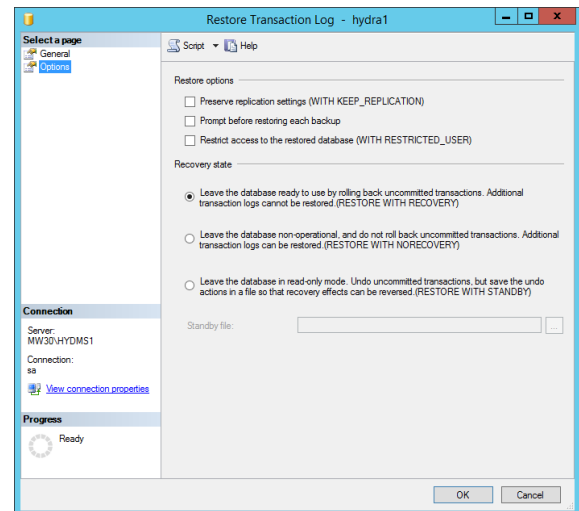
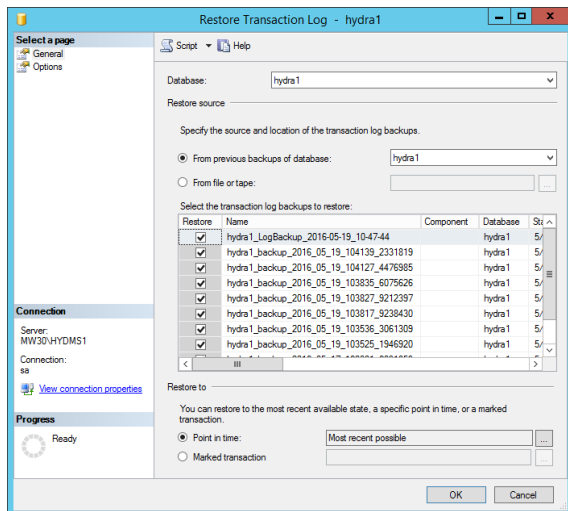


Now the existing transaction log backup of the database is loaded as an additional step.

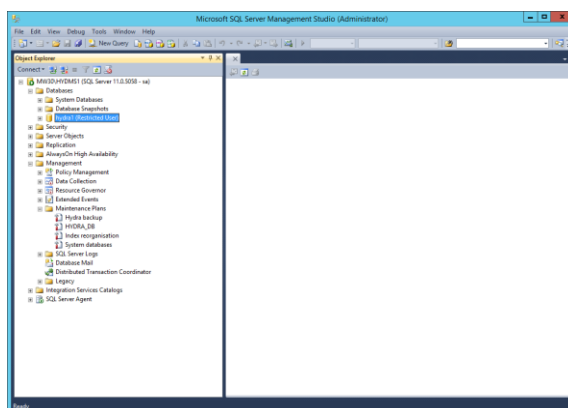
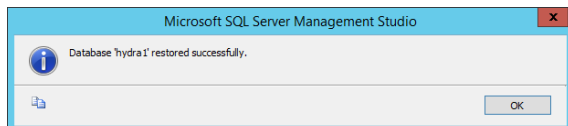
SERVERNAME → Databases → HYDRA1 → right mouse button → Tasks → Restore → Transaction log



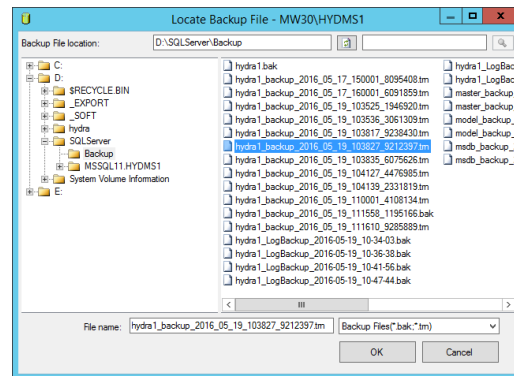
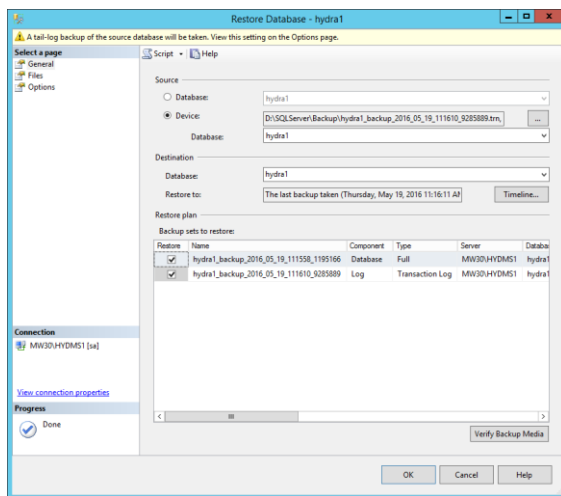
Select the menu item **Transaction log** in the field **Restore**. 'Transaction log' has to be selected in 'Backup record'.



After clicking on the OK button, the following screen opens (recovery progress). The duration of the database recovery depends on the size of the database.

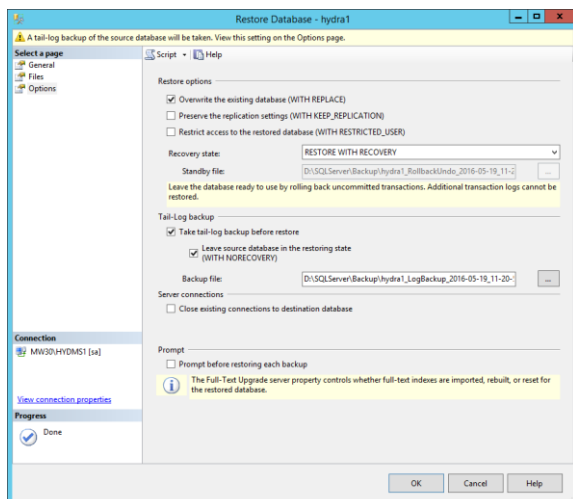


Restore from another medium



Selection of device

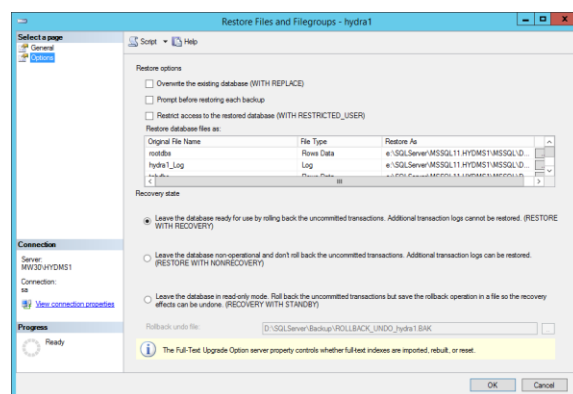
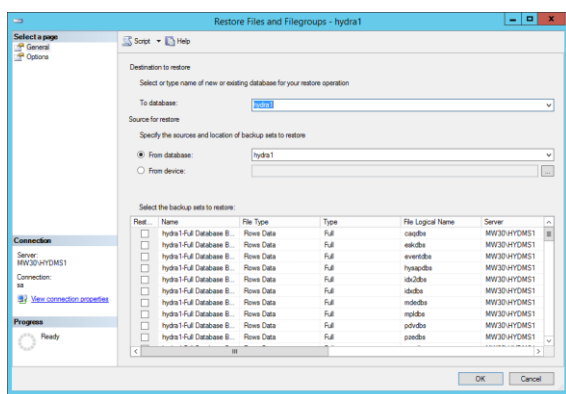
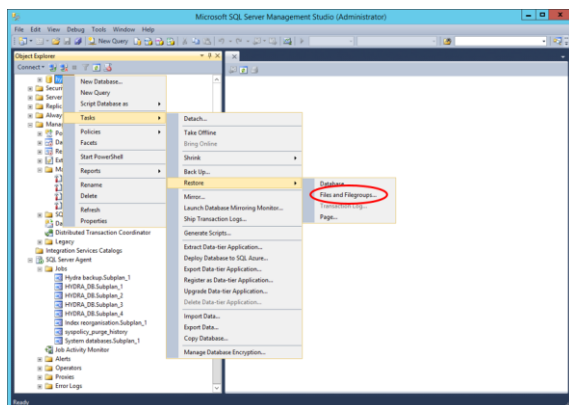
Indicate storage location of backup



The options must be set even if indicated by the medium

Restore individual files or file groups

SERVERNAME → Databases → HYDRA1 → right mouse button → Tasks → Restore → Files or file groups



SQL Server now shows the current files known to the system. In the given example, these is a complete **backup of all table spaces** of the HYDRA1 database. The file to be restored (e.g. CAQDBS) may be indicated here now.

In this example, the database recovery is indicated with the option **Latest possible time**. Depending on the error case, another option has to be indicated here (e.g. time-based or date-related).

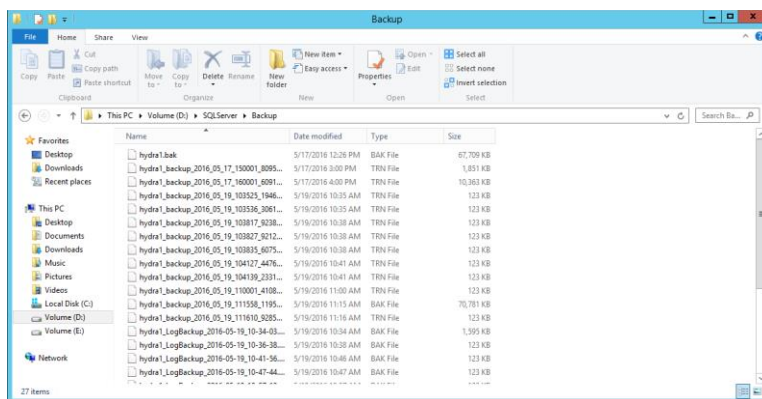
Restore on a new SQL Server installation

It is possible that a database is to be loaded to another hardware, e.g. after a hard drive crash. The problem with this is that the drives or directories may not have the same structure on the new host. If the directory structure is indicated when loading the database, the database may also be loaded on new hardware.

Example: Backup file from the existing productive server **hydra1.bak**

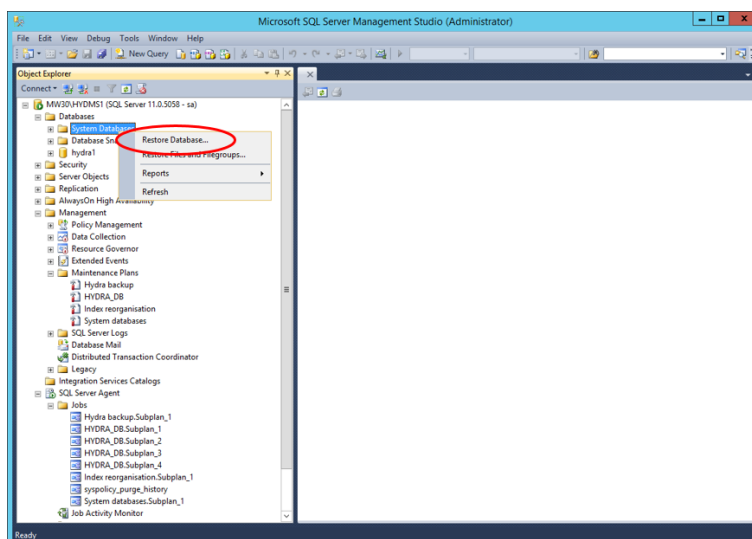
Step 1: Copy backup file to a new server

In the first step, the backup file is copied to a drive of the new server.

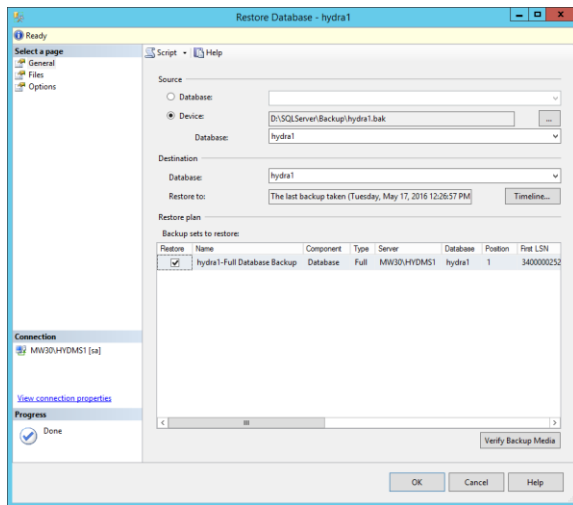


Step 2: Restore database

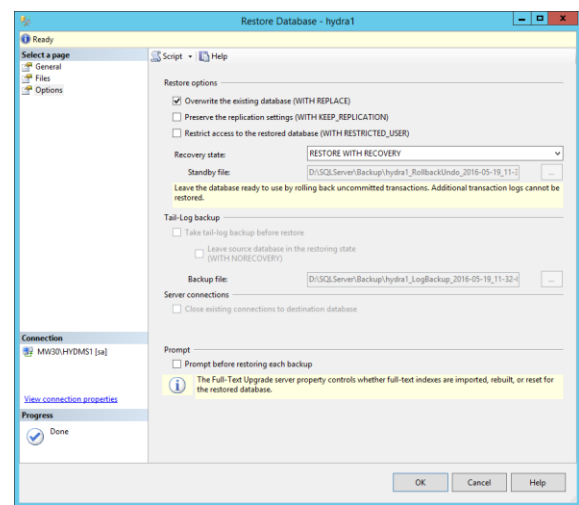
In the next step, the database is restored with SQL Server Management Studio.



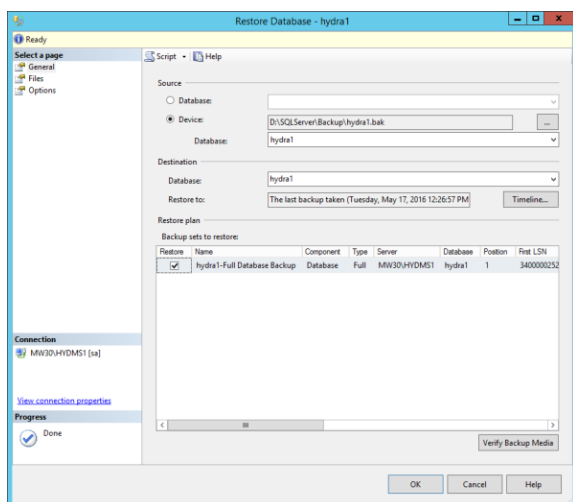
SERVERNAME → Databases → right mouse button → Restore database



Database: *hydra1*
Indicate source *From Medium*

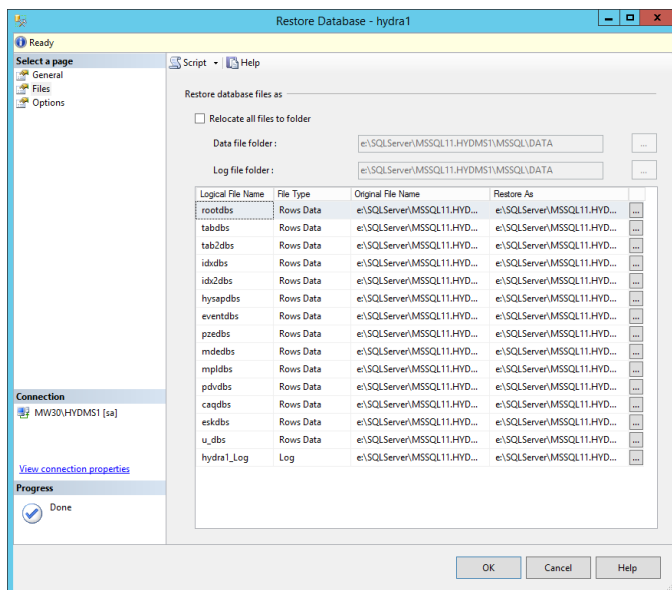


OK



Select backup file: *e.g. hydra1.bak*

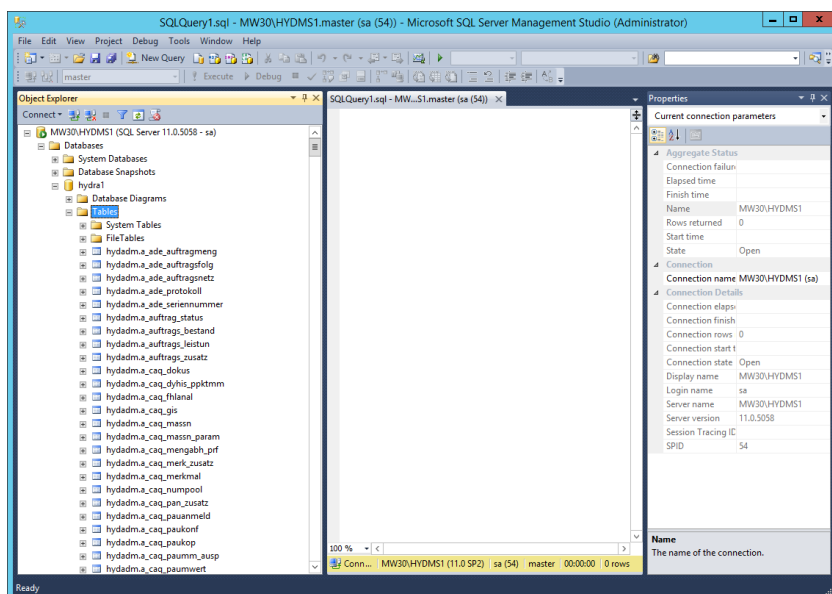
The indicated data record can now be selected.



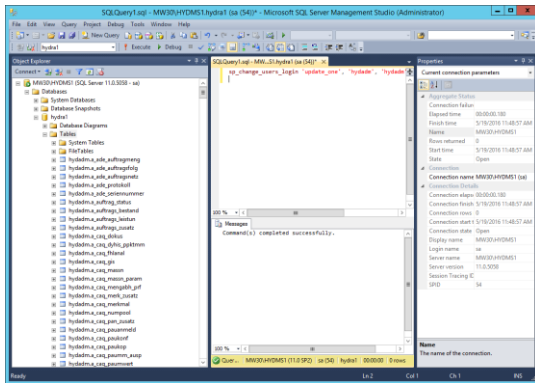
All file names can now be indicated manually in the Files tab. The directory structure must exist, the files themselves do not have to exist. (e.g. file name: tabdbs → Restore as E:\SQLServer\MSSQL11.HYDMS1\MSSQL\DATA\tabdbs_Data.NDF)

Step 3: Set rights for user hydadm

After loading the hydra1 database via the backup file, the database user hydadm must be assigned to an SQL Server log on name via the saved procedure sp_change_users_login. This is achieved by the following command:

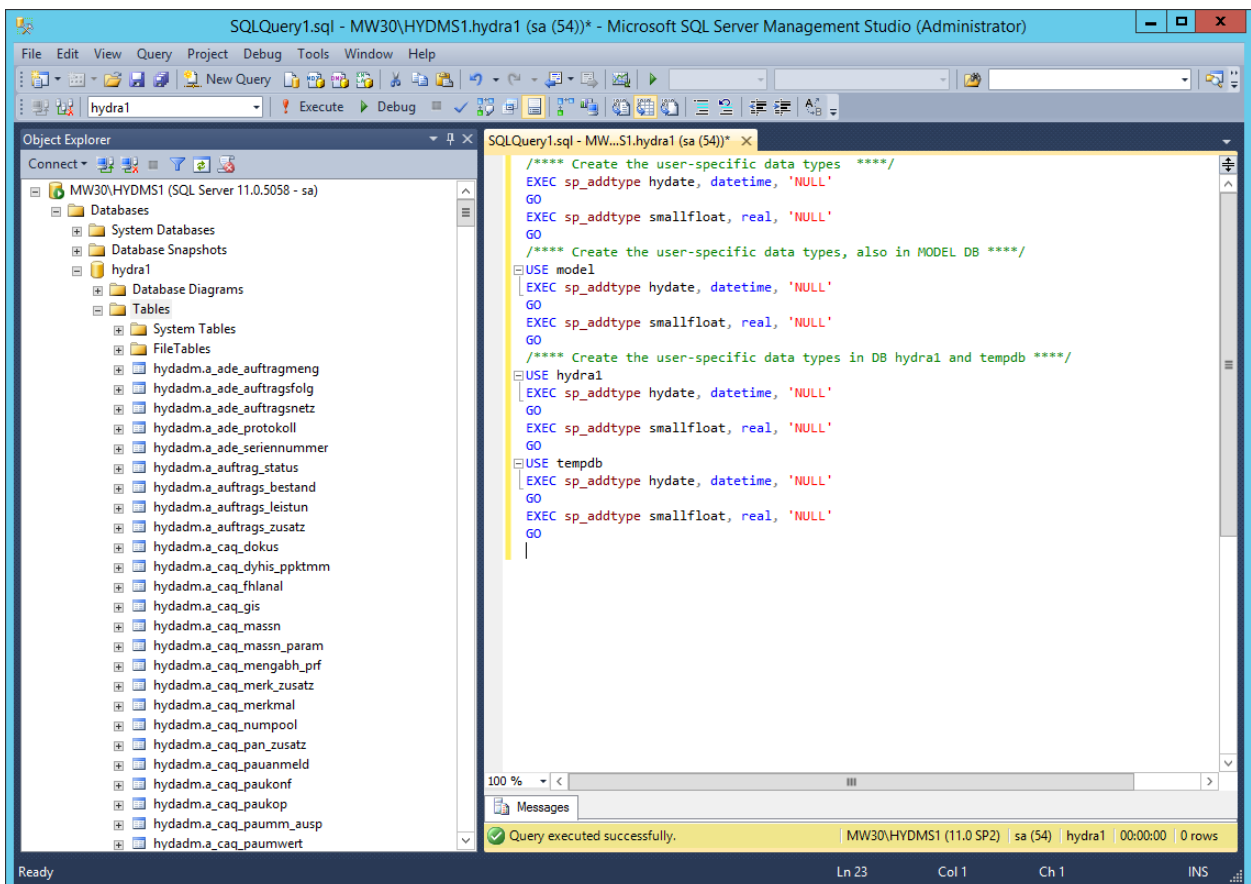


```
sp_change_users_login 'update_one', 'hydadm', 'hydadm'
```



Step 4: Create the user-specific data types

The user-specific data types generated when the HYDRA database is created must now be renewed in a separate step.



Copy template:

```

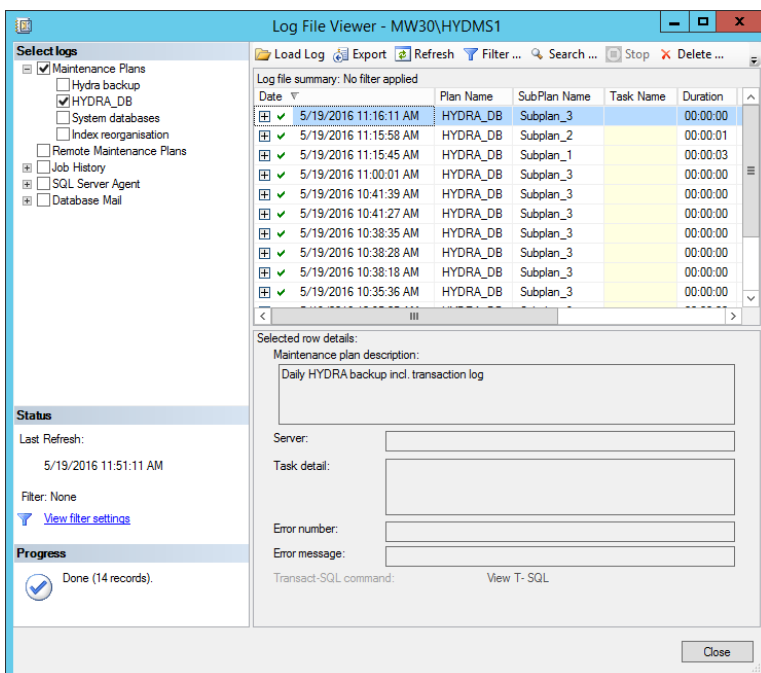
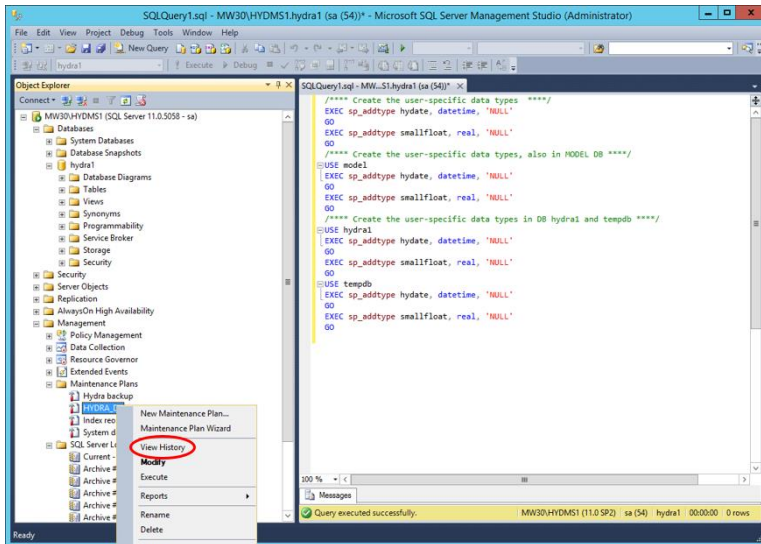
/**** Create the user-specific data types ****/
EXEC sp_addtype hydate, datetime, 'NULL'
GO
EXEC sp_addtype smallfloat, real, 'NULL'
GO
/**** Create the user-specific data types, also in MODEL DB ****/

```

```
USE model
EXEC sp_addtype hydate, datetime, 'NULL'
GO
EXEC sp_addtype smallfloat, real, 'NULL'
GO
/**** Create the user-specific data types in DB hydra1 and tempdb ****/
USE hydra1
EXEC sp_addtype hydate, datetime, 'NULL'
GO
EXEC sp_addtype smallfloat, real, 'NULL'
GO
USE tempdb
EXEC sp_addtype hydate, datetime, 'NULL'
GO
EXEC sp_addtype smallfloat, real, 'NULL'
GO
```


Show process of a maintenance plan

SERVERNAME → Administration → Maintenance plans → right mouse button → View history



Protokolldatei-Viewer - WIN2008-TEST\HYDMS

Protokolle auswählen

- ☐ Auftragsverlauf
- ☐ Datenbank-E-Mail
- ☐ Remotewartungspläne
- ☐ SQL Server-Agent
- ☒ Wartungspläne
 - ☐ HYDRA Backup
 - ☐ SYSTEMDATENBANKE
 - ☐ Index Reorganisaion
 - ☒ HYDRA_DB_MIT_TRP

Status

Letzte Aktualisierung:
26.08.2009 11:29:01

Filter: 'Keiner'

[Filtereinstellungen anzeigen](#)

Status

☒ Fertig (3 Datensätze).

Protokoll laden Exportieren Aktualisieren Filtern... Suchen... Löschen... Hilfe

Protokolldateizusammenfassung: 'Kein Filter angewendet'

Datum	Planname	Unterplanname	Dauer	Protokolltyp
26.08.2009 11:22:09	HYDRA_DB_MIT_TRP	Subplan_1	00:00:01	Wartungsplan
25.08.2009 22:00:02	HYDRA_DB_MIT_TRP	Subplan_1	00:00:01	Wartungsplan
25.08.2009 17:05:52	HYDRA_DB_MIT_TRP	Subplan_1	00:00:01	Wartungsplan

Details für die ausgewählte Zeile:

Wartungsplanbeschreibung:
HYDRA Datenbanksicherung (vollständig) incl. Transaktionsprotokoll

Server:

Taskdetails:

Fehlernummer:

Fehlermeldung:

Transact-SQL-Befehl: T-SQL anzeigen

Schließen